

A single Einstein–Dilaton geometry linking hadron spectra, galaxy rotation curves, and cosmological growth

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Abstract

A corrected, geometry-preserving formulation is given for a frozen five-dimensional Einstein–dilaton background used as a common computational input for hadronic, galactic, cosmological, and laboratory readouts. A three-stage audit first derives and seals the field equations, then compares the frozen trace with the previously declared canonical polynomial action, and finally adjudicates the result in an independent coordinate representation. The stored trace does not solve that polynomial system as labelled. It is shown, however, that the achieved warp and scalar profiles admit a healthy effective completion $S_{\text{eff}} \propto \int \sqrt{-g} (R - K(\phi)(\partial\phi)^2/2 - V(\phi))$. The reconstructed kinetic function is strictly positive on the certified radial interval, so the completion is equivalent to a canonical scalar by field redefinition. The warp and scalar profiles are preserved to 2.04×10^{-6} and 8.54×10^{-8} , respectively, while the complete field equations close at normalized floating-point residuals below 8×10^{-17} . The deformation used by the observational arms is recovered with correlation 0.999999905 and rms difference 6.88×10^{-4} . On a declared compact interval, the carrier gives $\beta_n = \sqrt{I_g/3} f_n$. Neumann endpoints yield $\beta_0 = 0.0542901$ and conditional force fraction $2\beta_0^2 = 5.89483 \times 10^{-3}$, but that massless unscreened branch violates Solar-System bounds. The NN/ND/DN/DD audit finds either the same zero mode, a UV-coupled ultralight mode, or point-probe decoupling; no boundary is selected. The six positive modes remain a dimensionless template with unfixed length. All stronger observational and laboratory readings retain explicit interface or dictionary assumptions. The electromagnetic audit corrects a radial-gauge error and derives a gauge-invariant bulk-Maxwell measure, scalar–photon vertex, and photon KK comb; localization, source, scale, and atomic response remain unselected. A conditional functional-BPS completion instead leaves a flat bi-scalar moduli sector with a positive two-dimensional kinetic metric. Each endpoint matter selector has a covariant stationary direction with a nonzero even jet, but no common or dynamically selected direction exists and minimal matter couples through the inverse selector; a physical $q^2 Y$ vertex is not derived.

The separate stiff completion has seven positive residues with $\sum_n \alpha_n = 0.106765$. Recomputed SPARC inputs give test $\chi^2/\text{point} = 371.58$, versus 414.23 for baryons and 36.75 for the train-only empirical RAR; the geometry-matched finite-range scan reaches its long-range bound, so no scale or force law is identified. The RAR-derived nonlinear action is a consistency target, not a bulk prediction, and is not relabelled as a HOLO force law. Likewise, the DESI comparison and the P7 breathing kernel are diagnostics without preference, source occupation, clock anchor, or detection.

Key words: gravitation – holography – galaxies: kinematics and dynamics – cosmology: large-scale structure of Universe – methods: numerical

1 Introduction

Einstein–scalar domain walls provide a controlled language for holographic renormalization-group flows and, with an appropriate infrared completion, for confining Yang–Mills phenomenology (DeWolfe et al., 2000; Gürsoy et al., 2008, 2011). The project studied here uses one frozen bulk profile as a shared input for a gauge-invariant scalar fluctuation operator, a SPARC input audit, a cosmological-growth dictionary, and versioned single-arm response operators. That common dependency and the accompanying machine-readable artefacts are an operational achievement: every arm can be rerun against the same frozen arrays.

Reproducibility of a common input is not, by itself, a derivation of all interfaces from one action. The stronger claim requires two separate checks. First, the bulk arrays must solve the field equations associated with the declared action and gauge. Secondly, mappings from the holographic radius to a galaxy radius, cosmological redshift, or laboratory time must either follow from an explicit interface action or be identified as phenomenological dictionaries.

This revision performs those checks without discarding the achieved geometry. Section 2 derives the complete bulk equations. Section 3 records the three-stage consistency audit and identifies a preconditioning label mismatch. Section 4 reconstructs a positive effective kinetic function and potential that sup-

port the frozen profiles. Section 5 derives one explicitly conditional probe-matter interface without observational calibration. The remaining sections retain the established numerical readouts with corrected evidential labels. No simultaneous multi-arm inversion or laboratory detection is claimed.

2 Effective Einstein–dilaton background

Signature $(-, +, +, +, +)$ is used together with the effective action

$$S_{\text{eff}} = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} \left[R - \frac{1}{2} K(\phi)(\partial\phi)^2 - V(\phi) \right]. \quad (1)$$

The canonical model is the special case $K = 1$. Variation gives

$$R_{MN} = \frac{1}{2} K \partial_M \phi \partial_N \phi + \frac{1}{3} g_{MN} V, \quad (2)$$

$$K \nabla^2 \phi + \frac{1}{2} K_{,\phi} (\partial\phi)^2 = V_{,\phi}. \quad (3)$$

In domain-wall gauge,

$$ds^2 = e^{2A(u)} \eta_{\mu\nu} dx^\mu dx^\nu + du^2, \quad \phi = \phi(u), \quad (4)$$

these equations reduce to

$$A'' = -\frac{1}{6}K\phi'^2, \quad (5)$$

$$12A'^2 - \frac{1}{2}K\phi'^2 + V = 0, \quad (6)$$

$$K(\phi'' + 4A'\phi') + \frac{1}{2}K_{,\phi}\phi'^2 = V_{,\phi}. \quad (7)$$

Primes in Eqs. (5)–(7) denote d/du . The constraint alone is not a substitute for the two dynamical equations.

For comparison in conformal gauge, define $dz/du = e^{-A}$. The metric is $ds^2 = e^{2A(z)}(\eta_{\mu\nu}dx^\mu dx^\nu + dz^2)$ and the equations transform covariantly. Pure AdS has constant $A_u = \pm L^{-1}$ in domain-wall gauge but $A_z = -z^{-1}$ in the usual conformal orientation. These reference derivatives must not be mixed.

2.1 Constraint propagation and matter sources

For the canonical vacuum system, differentiating $H = 12A'^2 - \phi'^2/2 + V$ and inserting the dynamical equations gives $H' = 0$. If a density is added only through $A'' = -\phi'^2/6 - \kappa\rho$ while the vacuum constraint is retained, then

$$H' = -24\kappa A'\rho. \quad (8)$$

A consistent matter source therefore requires its action, conserved stress tensor, and corresponding constraint contribution. Equation (8) is the reason the old sourced polynomial interpretation cannot be retained unchanged.

3 Three-stage audit of the frozen trace

The trace contains 1999 samples over the stored coordinate range $[0.01, 2]$. The audit was separated into: (i) direct derivation and sealing of the equations and thresholds; (ii) comparison with hash-frozen candidate outputs; and (iii) independent integration and coordinate-gauge adjudication. Input hashes and thresholds are included in the public audit artefacts.

An independent positive control is provided by

$$W(\phi) = \frac{6}{L} + \frac{\phi^2}{2L}, \quad V = \frac{1}{2}W_{,\phi}^2 - \frac{1}{3}W^2, \quad (9)$$

which gives

$$V = -\frac{12}{L^2} - \frac{3\phi^2}{2L^2} - \frac{\phi^4}{12L^2} \quad (10)$$

and the exact flow

$$\phi = \phi_0 e^{(u-u_0)/L}, \quad A = A_0 - \frac{u-u_0}{L} - \frac{\phi^2 - \phi_0^2}{24}. \quad (11)$$

It has $m^2 L^2 = -3$, above the AdS_5 Breitenlohner–Freedman bound (Breitenlohner & Freedman, 1982). Independent second-order integration agrees with the exact solution to 4.92×10^{-14} ; the largest conformal-gauge residual is 3.55×10^{-15} .

Table 1 reports the same tests on the frozen trace. Both the literal stored polynomial sign and the intended BF-mass sign fail the sealed criteria. The previously reported maximum raw constraint residual, 1.81, is therefore not treated as a harmless diagnostic.

The cause is identifiable. The original numerical preconditioner rescales the state, evaluates an untransformed right-hand

Table 1: Equation-level audit of the frozen trace. Values are normalized residuals; all listed tests fail their sealed criterion.

Metric	Literal	BF sign	Criterion
ϕ kinematic rms	0.3139	0.3139	≤ 0.01
A kinematic rms	0.4615	0.4615	≤ 0.01
Constraint p95	0.03852	0.03851	≤ 0.001
Scalar EOM rms	0.3220	0.3543	≤ 0.01
Sourced warp EOM rms	0.8903	0.8903	≤ 0.01

side, and then reverses the scaling. Consequently the stored columns satisfy approximately

$$A_{,u} \simeq u(dA)_{\text{stored}}, \quad \phi_{,u} \simeq (d\phi)_{\text{stored}}/u, \quad (12)$$

rather than being direct derivative columns. This is a provenance and label error, not evidence that the smooth warp geometry itself must be discarded.

The Bessel-based holonomic proxy in the previous version remains useful as a compact fit, but it imposes only Eq. (6). Its normalized constraint residual is 1.48×10^{-16} whereas its scalar and warp residuals are 0.9546 and 0.9494. That construction is therefore moved to the status of a descriptive proxy and is not used as the bulk action.

4 Geometry-preserving effective completion

For monotonic $\phi(u)$ and $A''(u) < 0$, Eqs. (5) and (6) can be inverted:

$$K(\phi(u)) = -\frac{6A''(u)}{\phi'(u)^2}, \quad V(\phi(u)) = -3A''(u) - 12A'(u)^2. \quad (13)$$

The scalar equation then follows from the derivative of the constraint wherever $\phi' \neq 0$. This is a geometry-preserving inverse completion: it asks which action supports the achieved profiles instead of replacing those profiles with the audit control solution.

Quintic smoothing splines remove sub-micro-scale integration noise. Ten samples at each endpoint are excluded because high-order derivatives are boundary sensitive. The certified interval is

$$0.010233 \leq u \leq 1.980981. \quad (14)$$

On this interval, the fit changes A by at most 2.04×10^{-6} and ϕ by at most 8.54×10^{-8} . The kinetic function obeys

$$8.55 \leq K(\phi) \leq 9.39 \times 10^6, \quad (15)$$

and is strictly positive at every certified sample. The wide range signals a poor normalization of the stored scalar near the ultraviolet endpoint, but not a kinetic ghost.

Defining a canonical scalar by

$$\chi'(u) = \sqrt{-6A''(u)} \quad (16)$$

gives an equivalent canonical representation. Table 2 shows closure in both field variables.

The operational deformation is also recovered. Equation (12) turns the stored expression $-u(dA)_{\text{stored}} - 1$ into the domain-wall quantity

$$\delta_{\text{eff}}(u) = -A'(u) - 1 \quad (L = 1). \quad (17)$$

Its correlation with the stored deformation is 0.999999905, with rms difference 6.88×10^{-4} . The shared observational input is therefore preserved to high precision.

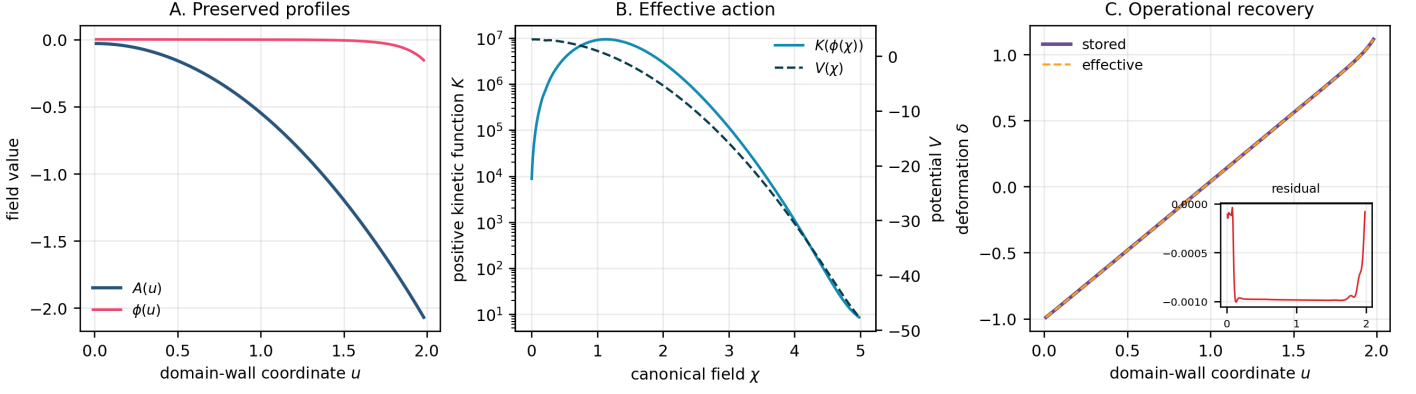


Figure 1: Geometry-preserving effective completion. Left: reconstructed warp and scalar profiles on the certified interval. Centre: positive kinetic function and potential in canonical field coordinates. Right: stored and reconstructed operational deformations and their residual.

Table 2: Certificate for the effective completion.

Quantity	Value
Non-canonical warp max residual	2.22×10^{-16}
Constraint max residual	7.11×10^{-15}
Non-canonical scalar normalized rms	7.58×10^{-17}
Canonical warp max residual	4.44×10^{-16}
Canonical scalar max residual	1.07×10^{-14}

5 Conditional compact-interval matter interface

The corrected bulk fixes a healthy local scalar carrier, but a holographic radial coordinate is not automatically an extra physical dimension. A blind forward benchmark is obtained by adding six explicit assumptions: one physical copy of the certified interval; a consistent Gibbons–Hawking–York/background/junction completion with no additional localized quadratic trace term; selected Neumann endpoints; probe Standard Model matter localized on a fixed slice; no brane bending or backreaction; and induced-metric coupling only at tree level. These assumptions are not selected by the frozen trace.

Let $u_{\text{phys}} = \ell u$ and define

$$\epsilon = -\frac{A''}{A'^2} > 0, \quad p = e^{4A}\epsilon, \quad w = e^{2A}\epsilon, \quad (18)$$

where the ratio defining ϵ is invariant under the constant rescaling from u to u_{phys} . The local transverse-trace action, with the normalization mapped from Arutyunov et al. (2000), is

$$S_h^{(2)} = -\frac{3\ell}{128\kappa_5^2} \int d^4x du \left[w \eta^{\mu\nu} \partial_\mu h \partial_\nu h + \frac{p}{\ell^2} (h')^2 \right]. \quad (19)$$

That calculation supplies the local quadratic normalization; it does not supply the compactification or the endpoint conditions used here.

With $p f_n'|_{UV,IR} = 0$, the dimensionless modes obey

$$-(p f_n')' = \mu_n^2 w f_n, \quad \int du w f_n f_m = \delta_{nm}, \quad (20)$$

$$h(x, u) = \sum_n q_n(x) f_n(u), \quad \varphi_n = \frac{\sqrt{3\ell}}{8\kappa_5} q_n. \quad (21)$$

The physical trace-carrier masses are $M_n^{(h)} = \mu_n/\ell$. This tower is not an additional scalar tower independent of the ζ operator. To see this, introduce the conformal coordinate $dz_c = e^{-A} du$ and $s = e^{3A}\epsilon$. Equation (20) becomes

$$-(s f_{n,z_c})_{,z_c} = \mu_n^2 s f_n. \quad (22)$$

Writing $\Psi_n = e^B f_n$ with

$$B = \frac{1}{2} \log s = \frac{3}{2} A + \frac{1}{2} \log \epsilon = \frac{3}{2} A + \log |X| + \text{const}, \quad X = \frac{\chi'}{3A'}, \quad (23)$$

gives

$$[-\partial_{z_c}^2 + B'^2 + B''] \Psi_n = \mu_n^2 \Psi_n. \quad (24)$$

This is the local gauge-invariant scalar operator. The present Neumann compactification, the stored Dirichlet–Neumann calculation, and their use of canonical χ versus stored ϕ do not impose the same physical boundary problem, so their numerical spectra need not agree. They nevertheless describe one scalar degree of freedom and cannot be double counted. In the holographic reading u is an RG coordinate and the poles are composite; in the compact reading it is a physical interval and the modes are four-dimensional scalars.

For a conserved probe stress tensor, the longitudinal trace-sector field decouples and the transverse trace contributes

$$\delta S_m = \int d^4x \sqrt{-g} \frac{hT}{8} = \sum_n \int d^4x \sqrt{-g} \frac{\beta_n}{M_{\text{Pl}}} \varphi_n T. \quad (25)$$

Tensor reduction on the same interval gives

$$M_{\text{Pl}}^2 = \frac{\ell I_g}{\kappa_5^2}, \quad I_g = \int du e^{2A}, \quad (26)$$

and therefore

$$\beta_n(u_m) = \sqrt{\frac{I_g}{3}} f_n(u_m). \quad (27)$$

The unknown κ_5 and the interval length cancel from the dimensionless coupling, although ℓ remains necessary to fix every nonzero force range.

The blind numerical reduction gives $I_g = 0.8079601$ and $I_w = 91.37501$. The Neumann zero mode is $f_0 = I_w^{-1/2}$, so

$$\beta_0 = \sqrt{\frac{I_g}{3I_w}} = 0.0542901, \quad 2\beta_0^2 = 5.89483 \times 10^{-3}. \quad (28)$$

If this massless branch survives the boundary completion, two unscreened nonrelativistic bodies have

$$V_{12}(r) = -\frac{G_N m_1 m_2}{r} \left(1 + 2\beta_0^2\right). \quad (29)$$

Thus the scalar term is 0.589 per cent of the tensor Newtonian term in this benchmark. Under the standard universal massless scalar-tensor mapping it would give

$$\gamma = \frac{1 - 2\beta_0^2}{1 + 2\beta_0^2} = 0.988279, \quad (30)$$

whereas Cassini found $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti et al., 2003). The massless unscreened Neumann branch is therefore excluded. A different boundary sector may lift or remove the zero mode, but it must be specified prospectively and its complete spectrum recomputed.

The computed dimensionless tower is

$$(\mu_0, \dots, \mu_3) = (0, 0.913899, 1.541695, 2.133778), \quad (31)$$

$$(\mu_4, \dots, \mu_6) = (2.728899, 3.336654, 3.954473), \quad (32)$$

with UV-slice couplings

$$(\beta_0, \beta_1, \beta_2) = (0.0542901, 0.000705106, 0.00155651), \quad (33)$$

$$(\beta_3, \beta_4) = (0.00233763, 0.00282007), \quad (34)$$

$$(\beta_5, \beta_6) = (0.00306861, 0.00320280). \quad (35)$$

The finite-element result passes mesh, orthonormality, residual, and zero-mode checks; a separate shooting solve provides the independent numerical check.

The six positive modes of the Neumann trace basis define the historical P6 benchmark, with $\sum \alpha_n = 7.20230 \times 10^{-5}$. They are not transplanted into the microscopic boundary problem. The quadratic gauge-invariant scalar action and its UV matter vertex instead canonically normalize the declared stiff-boundary candidate. Its seven masses and force residues are

$$(\mu_0, \dots, \mu_3) = (0.290482, 1.224178, 1.725447, 2.188318), \quad (36)$$

$$(\mu_4, \dots, \mu_6) = (2.744425, 3.344891, 3.961129), \quad (36)$$

$$(\alpha_0, \dots, \alpha_2) = (0.00101753, 0.00708333, 0.0314763), \quad (37)$$

$$(\alpha_3, \alpha_4) = (0.0316268, 0.0176374), \quad (37)$$

$$(\alpha_5, \alpha_6) = (0.0107024, 0.00722131). \quad (37)$$

Thus $\sum_n \alpha_n = 0.106765$ and the maximum circular-speed multiplier is $\sqrt{1 + \sum \alpha_n} = 1.052029$. The correlated mass ratios are

$$1 : 4.214303 : 5.939951 : 7.533412 : \\ 9.447842 : 11.514983 : 13.636417. \quad (38)$$

For $x = r/\ell$ they define the current material-force templates

$$\frac{a_\varphi}{a_N} = \sum_{n=0}^6 \alpha_n (1 + \mu_n x) e^{-\mu_n x}, \quad (39)$$

$$\frac{|\partial_r a_\varphi|}{|\partial_r a_N|} = \sum_{n=0}^6 \alpha_n \left(1 + \mu_n x + \frac{\mu_n^2 x^2}{2}\right) e^{-\mu_n x}. \quad (40)$$

A mechanical mode with angular frequency ω_m and quality factor Q then responds to a source-modulated differential acceleration as

$$|\Delta x(\Omega)| = \frac{|\Delta a(\Omega)|}{\sqrt{(\omega_m^2 - \Omega^2)^2 + (\Omega \omega_m / Q)^2}}. \quad (41)$$

Equations (40)–(41) are a transfer law, not a predicted signal: a source trajectory, ℓ , detector geometry, elastic response, noise, and mode occupation must still be supplied.

The stiff force in Eq. (37) is specifically the static, $\Omega = 0$ slice. Restoring the four-dimensional dynamics of each positive scalar mode gives

$$\left(\partial_t^2 - c^2 \nabla^2 + \omega_n^2\right) \varphi_n = J_n, \quad \omega_n = \frac{c \mu_n}{\ell}. \quad (42)$$

For a harmonic source $J_n \propto e^{-i\Omega t}$ and $v = \Omega \ell / c$, the outgoing radial Green function is

$$G_n(r, \Omega) \propto \frac{1}{r} \begin{cases} \exp[-r\sqrt{\mu_n^2 - v^2}/\ell], & v < \mu_n, \\ \exp[+ir\sqrt{v^2 - \mu_n^2}/\ell], & v > \mu_n. \end{cases} \quad (43)$$

Thus the static Yukawa inverse range is reduced as a source approaches the rest frequency; above threshold the response propagates with $v_g/c = \sqrt{1 - (\omega_n/\Omega)^2}$. We call this observation-free dynamic completion P7. It recovers every stiff-force value at $\Omega = 0$ to machine precision but is not an amplitude enhancement inserted into SPARC: a static galaxy has no source power at the required frequency.

If an independently measured cyclic frequency f_0 were prospectively identified with the first stiff mode, it would fix

$$\ell = \frac{\mu_0 c}{2\pi f_0}, \quad \frac{f_n}{f_0} = \frac{\mu_n}{\mu_0}, \\ \left(\frac{f_0}{f_0}, \dots, \frac{f_6}{f_0}\right) = (1, 4.214303, 5.939951, 7.533412, \\ 9.447842, 11.514983, 13.636417). \quad (44)$$

The first-mode static-range light-crossing time would then be $\lambda_0/c = (2\pi f_0)^{-1} = T_0/(2\pi)$. A rectangular coherent record needs at least $0.628T_0$ merely to place the first spectral zeros between all adjacent teeth of this comb. That is not a detection time: the signal front cannot arrive before r/c , a propagating packet requires approximately r/v_g , and a damped resonator has ring-up time $Q/(\pi f)$. Source occupation, phase, coherence or linewidth, boundary action, and detector noise remain external physical inputs.

For a Maxwell field localized on a four-dimensional slice, $T_{EM} = 0$ and classical Weyl invariance gives no independent local $\varphi_n F_{\mu\nu} F^{\mu\nu}$ vertex from the conformal induced metric. This statement does not apply to a photon propagating through the fifth dimension: its action also samples the perturbed radial lapse, whose constraint-generated coupling is derived below. A universal rescaling of all masses does not by itself change leading clock-frequency ratios, and anomaly or non-universal coefficients remain separate possible channels (Damour & Donoghue, 2010).

6 Data and evaluation protocol

Table 3 records unaffected numerical cross-checks against the original PDF together with the current SPARC adjudication. The galaxy comparators are baryons alone and the empirical RAR under the same evaluation protocol; this is not a benchmark against state-of-the-art dark-matter fits.

6.1 SPARC physical-input contract

SPARC provides signed gas contributions and resolved rotation curves for 175 galaxies with $3.6 \mu\text{m}$ photometry (Lelli et al.,

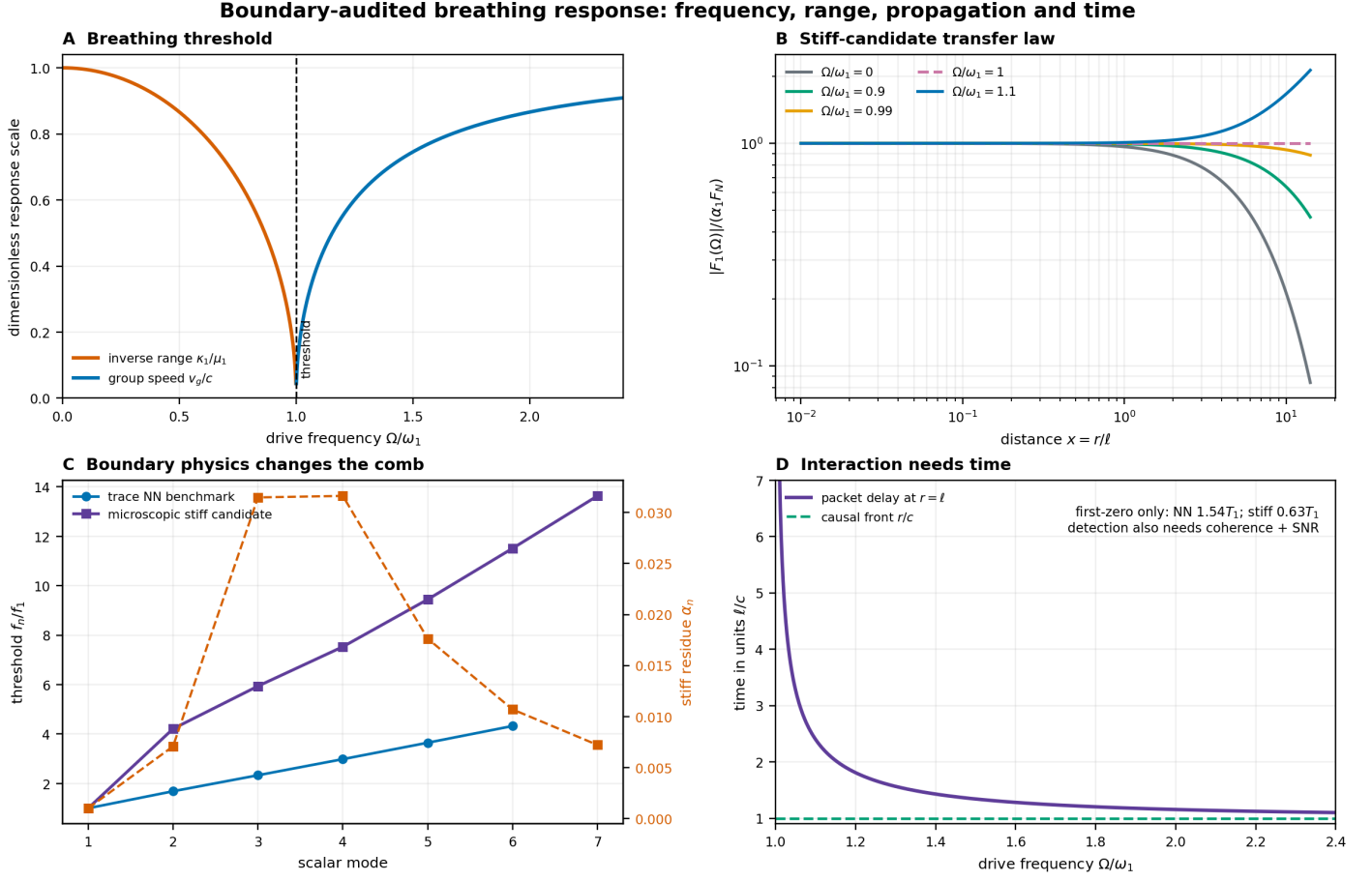


Figure 2: P7 conditional breathing response. (A) Below the first rest frequency the inverse Yukawa range collapses toward the threshold; above it an outgoing massive packet has subluminal group speed. (B) The stiff force is the static slice, while time-dependent drives change the radial transfer. (C) The seven stiff thresholds and canonically normalized residues are shown against the old six-mode NN trace benchmark. (D) Causal and group delays show why a frequency match alone is insufficient. All axes are dimensionless; no historical mHz value or observational residual is used.

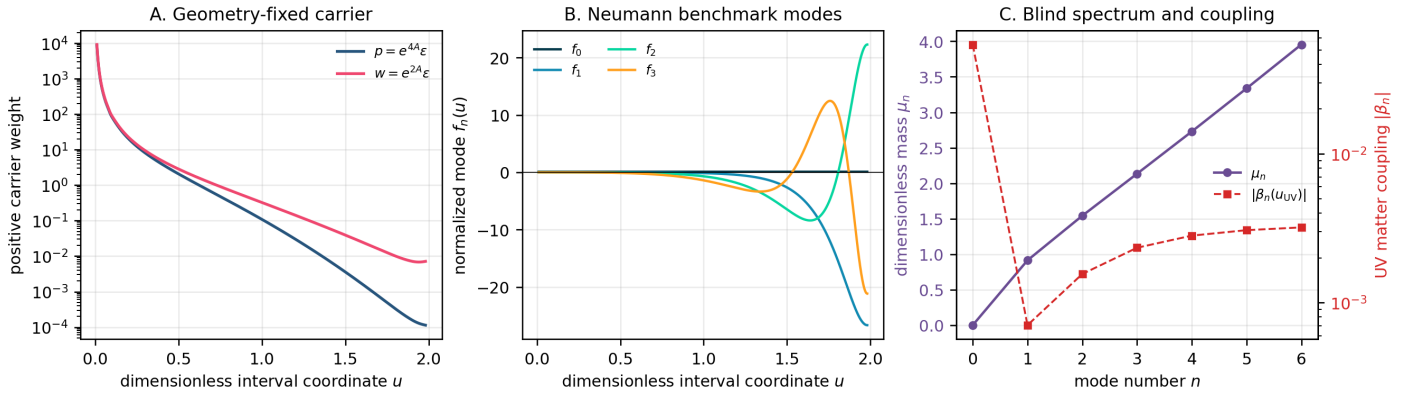


Figure 3: Blind conditional matter completion. Left: positive carrier weights fixed by the reconstructed geometry. Centre: normalized Neumann benchmark modes. Right: dimensionless trace-carrier masses and UV-slice matter couplings. No observational residual or historical fitted coupling enters this calculation.

Table 3: Current numerical status. Dict/ Λ CDM denotes the cosmological dictionary model and its matched flat- Λ CDM reference.

Quantity	Original PDF	Revision	Evidential status
Stored scalar ratio	1.5455	1.5454665	operator diagnostic
IR scale proxy	0.203 GeV ² ; 451 MeV; 1.60 GeV	unchanged	external normalization finite-disk scan reaches
SPARC test χ^2 /point (stiff/RAR)	—	371.58/36.75	long-range boundary and misses
Historical P5 comparator	—	290.98	provenance only
“600” audit (radius cut/range)	—	390.85/371.72	stiff force still rejected conditional no-go;
QCD/galaxy ℓ ratio	—	8.61×10^{40}	neither scale measured
BOSS covariance χ^2 (dict/ Λ CDM)	2.266/2.443	unchanged	statistically indistinguishable
High- z growth shift	≈ -12 per cent	unchanged	composite-dictionary prediction

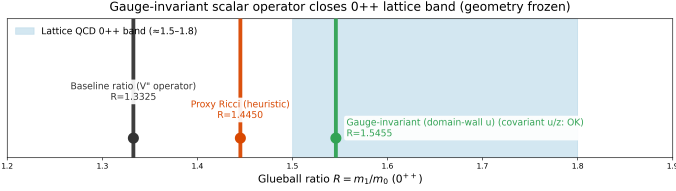


Figure 4: Stored scalar 0^{++} operator ratio on the frozen profile. In this revision $m_1/m_0 = 1.5455$ is retained as a reproducible operator diagnostic; it is not yet promoted to a physical glueball prediction of the reconstructed effective action.

2016). The audit uses

$$v_{\text{bar}}^2 = v_{\text{gas}} |v_{\text{gas}}| + 0.5v_{\text{disk}}^2 + 0.7v_{\text{bul}}^2, \quad (45)$$

with negative totals clipped only before taking the square root. All 3391 catalogue points are recomputed. A deterministic 122/26/27 galaxy split is used for train, validation, and test; no test velocity enters parameter fitting or curve selection.

6.2 Growth and BOSS DR12

The cosmological arm uses the explicit dictionary

$$z_{\text{trace}} = 1 + z_{\text{cos}}, \quad N = -\ln z_{\text{trace}}. \quad (46)$$

It is not derived from a compactification or a four-dimensional FRW action. Predictions that use Eq. (46) are consequently labelled dictionary-dependent. The comparison uses the BOSS DR12 covariance at $z = \{0.38, 0.51, 0.61\}$ (Alam et al., 2017) and a matched Planck-2018 flat- Λ CDM reference with $\Omega_{m0} = 0.315$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and $\sigma_{8,0} = 0.811$ (Planck Collaboration, 2020).

7 Gauge-invariant scalar operator

For a self-adjoint fluctuation problem,

$$-(p\psi')' + q\psi = \lambda w\psi, \quad (47)$$

normalizable solutions with declared ultraviolet and infrared boundary conditions define a discrete finite-domain spectrum. The stored scalar channel uses a Schrödinger potential $V_\zeta = B'^2 + B''$, with $B = 3A/2 + \log X$, Dirichlet ultraviolet and Neumann infrared conditions.

The operator calculation gives

$$(m_1/m_0)_{\text{stored}} = 1.5454665 \quad (48)$$

and agrees between its two covariantly transformed numerical realizations at the 10^{-7} level. This establishes internal coordinate consistency of the stored operator calculation. A physical glueball claim additionally requires recomputation using the canonical field χ , a confining infrared completion, normalizability, and prospectively fixed boundary conditions (Gürsoy et al., 2008, 2011; Morningstar & Peardon, 1999). The stored value is compatible with the broad early estimate 1.54(11) of Morningstar & Peardon (1999), but the modern continuum SU(3) values quoted by Athenodorou & Teper (2020) imply $m_{0^{++}}/m_{0^{++}} = 1.7195(160)$. The stored ratio is about 10.1 per cent lower and is not a sub-percent lattice prediction.

7.1 Infrared scale proxy

The stored quantity

$$\sigma_{\text{proxy}} = \frac{e^{2A(u_{\text{IR}})}}{2\pi\alpha'} \quad (49)$$

uses the finite endpoint rather than a demonstrated world-sheet turning point. With $e^{2A(u_{\text{IR}})} = 1.471 \times 10^{-2}$ and the externally chosen $\alpha' = 1.1527 \times 10^{-2} \text{ GeV}^{-2}$, it gives $\sigma_{\text{proxy}} = 0.203 \text{ GeV}^2$ and $\sqrt{\sigma_{\text{proxy}}} = 451 \text{ MeV}$. Applying the external scalar-channel factor $c = 3.55$ gives $m_0 \approx 1.60 \text{ GeV}$. These are model-dependent scale conversions, not parameter-free predictions. The actual causal direction is therefore frozen ED endpoint plus external α' and c , followed by σ_{proxy} and m_0 ; no Wilson loop feeds the geometry. The linked lattice-gauge artefact currently supplies elementary plaquettes and a plaquette correlator, not rectangular $W(R, T)$, a static potential, or a Creutz area-law extraction. Moreover a true holographic loop requires the string frame. Identifying the reconstructed χ with a standard IHQCD dilaton would give $A_s = A \pm \chi/\sqrt{6}$; neither orientation has a smooth interior minimum on the certified interval. An endpoint tension therefore additionally assumes a hard wall. Even a future lattice result $a^2\sigma$ fixes ℓ only if an independent matching relation ℓ/a is derived.

The numerical incompatibility is already decisive if one nevertheless forces a single scale onto both present interpretations. Identifying the first stiff mode, $\mu_0 = 0.290482$, with the legacy 1.600006 GeV proxy would require

$$\ell_{\text{QCD}} = \frac{\hbar c \mu_0}{1.600006 \text{ GeV}} = 3.58248 \times 10^{-17} \text{ m}, \quad (50)$$

and an interaction range $1.23329 \times 10^{-16} \text{ m}$. The SPARC scan instead saturates at its upper boundary $\ell_{\text{gal}} = 10^5 \text{ kpc}$, where the

first range is 3.44256×10^5 kpc and

$$f_0 = \frac{c\mu_0}{2\pi\ell_{\text{gal}}} = 4.49168 \times 10^{-18} \text{ Hz}, \quad T_0 = 7.05484 \text{ Gyr}. \quad (51)$$

Thus $\ell_{\text{gal}}/\ell_{\text{QCD}} = 8.61325 \times 10^{40}$. Neither number is a measurement—the QCD value is an external endpoint proxy and the galaxy value is a saturated grid edge—but one ℓ cannot realize both. A UV matching relation that separates effective scales, a distinct ultralight sector, or abandonment of one interpretation is compulsory.

8 SPARC physical input and empirical target

The deterministic galaxy split remains 122/26/27 for train, validation, and test. Only the training galaxies set the global acceleration scale. The empirical radial-acceleration relation (RAR) is

$$g_{\text{RAR}} = \frac{g_{\text{bar}}}{1 - \exp[-\sqrt{g_{\text{bar}}/g_{\dagger}}]}, \quad g_{\dagger} = 1.14414 \times 10^{-10} \text{ m s}^{-2}. \quad (52)$$

No observed test velocity is read to construct its curves. On all 621 test points, Eq. (52) gives $\chi^2/\text{point} = 36.75$ and a median absolute velocity error of 14.5 per cent. The large diagonal statistic is not hidden: distance, inclination, stellar mass-to-light uncertainty, intrinsic scatter, and covariance have not been marginalized here.

The old six-mode trace benchmark has $\sum_n \alpha_n = 7.20230 \times 10^{-5}$ and gives $\chi^2/\text{point} = 414.20$ in its exact long-range P6 limit. It is retained for numerical genealogy, not used as the current physical force. The canonically normalized stiff candidate in Eq. (37) instead has $\alpha_+ = 0.106765$ and therefore

$$V_{\text{stiff},\infty} = V_{\text{bar}}\sqrt{1 + \alpha_+}. \quad (53)$$

No SPARC datum enters the masses or residues. Equation (53) gives $\chi^2/\text{point} = 371.58$ on test: a material improvement over baryons alone at 414.23, but still an order of magnitude worse than the empirical RAR under the same diagonal statistic.

To test whether disk cancellation could change that conclusion, the finite range is applied to an effective axisymmetric thin disk. If $A_1(k) = \int R dR g_{\text{bar}}(R) J_1(kR)$, the geometry-matched forward operator is

$$g_{\text{stiff}}(R) = \int k dk A_1(k) J_1(kR) \left[1 + \sum_n \alpha_n \frac{k}{\sqrt{k^2 + (\mu_n/\ell)^2}} \right]. \quad (54)$$

A single global ℓ is selected on the 122 training galaxies over 10^{-3} – 10^5 kpc; there are no per-galaxy force parameters. The training objective decreases monotonically to the upper grid boundary, and the same result holds after doubling the spectral resolution and varying the outer V^2 continuation between $R^{-1/2}$ and $R^{-3/2}$. At the frozen boundary value the test score is 371.57906, numerically converged to the exact long-range result. Hence the finite disk does not select a scale or rescue the stiff candidate. This remains an effective-source check: the public SPARC mass-model archive provides the component rotation curves and stellar surface brightness, but not the gas surface-density map or a vertical-density profile, so it cannot identify a unique three-dimensional Yukawa convolution.

The origin-to-destination differential makes the missing shape explicit:

$$\Delta v(g_{\text{bar}}) = v_{\text{RAR}}(g_{\text{bar}}) - (1 + \alpha_+). \quad (55)$$

It changes sign at $g_{\text{bar}} = 6.25719 \times 10^{-10} \text{ m s}^{-2}$: the stiff limit oversupplies the small empirical correction at high acceleration and increasingly undersupplies it toward low acceleration. A fixed positive point-Yukawa comb is bounded by α_+ and its boost falls with radius; Eq. (54) shows that public-data disk nonlocality does not reverse the catalogue-level trend. The residual is therefore a matched-filter target for a new common microscopic source or state-dependent coupling, not a term to be fitted and added separately to each galaxy.

The corresponding universal signed collector can be written without any per-galaxy coefficient as

$$v_{\text{col}}(g_{\text{bar}}) = \left[1 - e^{-\sqrt{g_{\text{bar}}/g_{\dagger}}} \right]^{-1}, \quad \Delta g = [v_{\text{col}} - (1 + \alpha_+)] g_{\text{bar}}. \quad (56)$$

Only the 122 training galaxies determine $g_{\dagger}$; validation and test are frozen, and no object-specific residual is copied. Across the catalogue’s sampled domain, $0.08 \leq R \leq 108.31$ kpc and $2.09 \times 10^{-13} \leq g_{\text{bar}} \leq 7.34 \times 10^{-9} \text{ m s}^{-2}$, Eq. (56) requires $1.00033 \leq v_{\text{col}} \leq 23.9192$. By contrast, every positive linear Yukawa realization of the stiff spectrum obeys $1 \leq v_Y \leq 1 + \alpha_+ = 1.106765$: 94.3 per cent of catalogue points require more than that ceiling, while the high-acceleration 5.7 per cent require screening below it. Thus tuning a length cannot solve the mismatch. A future completion must derive a nonlinear/state-dependent equation or a genuinely distinct infrared sector; modified-Poisson constructions illustrate the type of action-level consistency test required, but are not adopted here (Milgrom, 2010).

The historical “600” statement has two possible readings and neither is a success. Keeping only test points with $R \geq 0.6$ kpc gives $\chi^2/\text{point} = 390.85$ for the stiff limit versus 38.30 for the collector. Instead setting the global Yukawa range to $\ell = 600$ kpc gives 371.72. These are galactic-domain checks, not evidence that one law works at all physical length scales. Equation (56) reaches 36.75 on the full test split precisely because it is the train-frozen empirical RAR target; it is not promoted to an action-derived HOLO prediction or a force detection.

8.1 Nonlinear action target and mass-dependent transition

Equation (56) can nevertheless define a nonrelativistic action target. Let

$$S_{\text{col}} = - \int d^3x \left[\frac{a_0^2}{8\pi G} \mathcal{F}(X) + \rho \Phi \right], \quad (57)$$

$$X = \frac{|\nabla \Phi|^2}{a_0^2}, \quad a_0 = g_{\dagger}.$$

Variation gives

$$\nabla \cdot [\mu(|\nabla \Phi|/a_0) \nabla \Phi] = 4\pi G \rho, \quad \mu(\sqrt{X}) = \mathcal{F}'(X). \quad (58)$$

In spherical symmetry, put $y = g_N/a_0$, $x = g/a_0$, and $t = \sqrt{y}$. Exact closure of the collector follows from the parametric map

$$\mu = 1 - e^{-t}, \quad x = \frac{t^2}{1 - e^{-t}}, \quad \mu(x)x = y. \quad (59)$$

Numerically integrating $\mathcal{F}'(X) = \mu$ gives $\mathcal{F} \sim (2/3)X^{3/2}$ in the deep regime and $\mathcal{F} \sim X$ in the Newtonian regime. Both principal coefficients of the field operator, μ and $\mu + x\mu'$, are positive for nonzero field but tend to zero in the deep limit. The equation is therefore degenerately, not uniformly, elliptic; no global existence or uniqueness theorem is claimed. A numerical inversion of the reconstructed constitutive table closes to 3.46×10^{-10} , and a spherical Plummer finite-difference control gives a maximum relative PDE residual of 1.26×10^{-5} . These are numerical consistency checks, not independent evidence. The construction belongs to the nonlinear action class introduced by [Bekenstein & Milgrom \(1984\)](#); a relativistic metric, lensing law, and external field treatment remain absent.

The current HOLO scalar sector fails a sharper embedding gate. Its positive quadratic carrier has $p_{\min} = 1.14 \times 10^{-4}$ and $w_{\min} = 6.83 \times 10^{-3}$, while the Neumann, positive-Robin, and stiff boundary choices alter masses and endpoint conditions but introduce no derivative kinetic function. Consequently, for a rescaled source $\rho \rightarrow \lambda\rho$, the regular weak-field branch obeys $g_{\text{HOLO}} = \lambda g_1 + O(\lambda^2)$. The collector instead requires $g_{\text{col}} \propto \sqrt{\lambda}$ in the deep limit, equivalently $\mathcal{F}(X) \propto X^{3/2}$ rather than X . A regular invertible field redefinition cannot change that source exponent. This is a conditional no-go for the present linearized canonical sector and its boundary-potential family; it does not exclude a new derivative operator, an IR critical branch, or another nonperturbative HOLO completion. Adding such an operator without deriving both its function and a_0 would be a new theory input, not a result of the existing bulk.

The scale is therefore an acceleration, not a fixed distance. For an isolated source,

$$r_M = \sqrt{\frac{GM}{a_0}}. \quad (60)$$

Thus $r_M = 0.6$ kpc implies $M = 2.955 \times 10^8 M_\odot$, while $r_M = 600$ kpc implies $M = 2.955 \times 10^{14} M_\odot$. A $6 \times 10^{10} M_\odot$ baryonic galaxy gives $r_M = 8.55$ kpc, and the Sun gives $r_M = 7199$ au. The formal isolated high-acceleration correction is exponentially negligible at Earth and planetary accelerations, but that is not a Solar-System validation because external fields and a relativistic completion have not been included.

An axisymmetric finite-volume control makes the nonspherical limitation explicit. In cylindrical coordinates the operator converges at second order on a Plummer benchmark: the fine-grid weighted relative L_2 residual is 2.21×10^{-3} and the coarse-to-fine ratio is 4.00. The algebraic field $\nu(|\mathbf{g}_N|/a_0)\mathbf{g}_N$ is curl-free for the spherical control, but a flattened Miyamoto–Nagai source gives normalized curl 2.11×10^{-2} ; it is not the gradient of an AQUAL potential.

A bounded physical-source prototype now combines the archived stellar surface-density profiles, Rotmod photometric anchors, a positive projected HI reconstruction normalized by M_{HI} and R_{HI} , spherical bulges, and monopole outer boundaries. The source is accepted only when a Newtonian forward calculation closes each component to a frozen 0.15 tolerance before V_{obs} is read. NGC 2403 and NGC 3198 pass, with total source closure 0.0501 and 0.0184; DDO 154 and NGC 2841 fail through their gas reconstructions. All four nonlinear solves converge to residuals near 3×10^{-6} , but only the two passing sources may be scored. They give respectively $\chi^2/\text{point} = 237.10$ and 10.20, with median fractional velocity errors 9.77 and 5.08 per cent. These are low-resolution diagnostics, not a catalogue validation: the global a_0 is inherited from the earlier V_{obs} -based training fit, and the gas inversion and vertical profiles remain modelling assumptions.

8.2 Where the nonlinear selector can and cannot couple

The exact Legendre envelope rewrites the exposed target as

$$\mathcal{F}(X) = \sup_{0 < s < 1} \{sX - W(s)\}, \quad W'(s) = X, \quad (61)$$

with $W(s) \sim s^3/3$ in the deep limit. This identity identifies the needed constitutive variable but not its microscopic carrier. A conventional three-dimensional soft bosonic determinant has the required cubic power but coefficient $-1/(12\pi)$ and leaves an analytic term proportional to X ; the present realization is therefore rejected. A filled gapped phase-space ball gives a positive cubic rest-energy term, but its natural thermodynamic conjugate is particle number rather than s , and neither its occupation nor normalization follows from the HOLO action.

The certified matter interface nevertheless exposes why the previous pole sums lost the relevant branch. In Einstein variables it contains $S_m[A_m(\varphi)^2 g_E, \Psi]$. Defining the matter metric and selector by

$$g_J = A_m^2 g_E, \quad s = A_m^{-2}, \quad (62)$$

gives the exact Jordan-frame terms

$$S_J \supset \int \sqrt{-g_J} \left[\frac{M_{\text{Pl}}^2}{2} s R_J - \frac{1}{2} Z_J(s) (\partial s)^2 - U_J(s) \right] + S_m[g_J, \Psi], \quad (63)$$

where $U_J = s^2 U_E$ and, for $\ln A_m = \beta\varphi/M_{\text{Pl}}$, $Z_J/M_{\text{Pl}}^2 = [1/(4\beta^2) - 3/2]/s$. This is an exact change of frame, not yet the auxiliary AQUAL action. The correctly normalized target would be

$$S_{\text{target}} = -M_{\text{Pl}}^2 \int d^3x [s|\nabla\Phi|^2 - a_0^2 W(s)] - \int d^3x \rho \Phi, \quad (64)$$

whose variations give $X = W'(s)$ and $\nabla \cdot (s\nabla\Phi) = 4\pi G\rho$. Equation (63) cannot be replaced by Eq. (64) before lapse, shift, the second metric potential, and scalar-gradient mixing are eliminated.

Moreover, the direct identification is obstructed in the isolated deep limit. With $t = \sqrt{g_N}/a_0$, the target has $s = 1 - e^{-t} \sim t$. Hence $A_m = s^{-1/2} \sim t^{-1/2}$ and the full Jordan tensor coefficient obeys $M_J^2/M_{\text{Pl}}^2 = s \rightarrow 0$. The frame map becomes singular precisely where the deep law is required. The surviving architecture therefore keeps the Einstein–Hilbert tensor term nondegenerate and places the selector in a distinct scalar or collective derivative sector, schematically

$$S_E \supset \int \sqrt{-g_E} \left[\frac{M_{\text{Pl}}^2}{2} R_E - M_{\text{Pl}}^2 a_0^2 \mathcal{F}(Y) \right] + S_m[A_m^2 g_E, \Psi]. \quad (65)$$

Its deep $\mathcal{F} \sim (2/3)Y^{3/2}$ can produce a scalar response proportional to $\sqrt{M}$ without switching off the tensor graviton. This operator, its a_0 , its causal covariant completion, and its lensing response have not yet been derived from the five-dimensional action. Expanding A_m and the scalar action around $s = 1$ before solving freezes the susceptibility and necessarily returns the mass-linear massless/Yukawa response; that early linearization is the concrete reason the nonlinear branch did not survive the earlier route.

The distinction can now be made quantitative. The first raw cubic vertex can already be isolated directly from the canonical

Nonlinear collector action target: one acceleration scale, mass-dependent radii

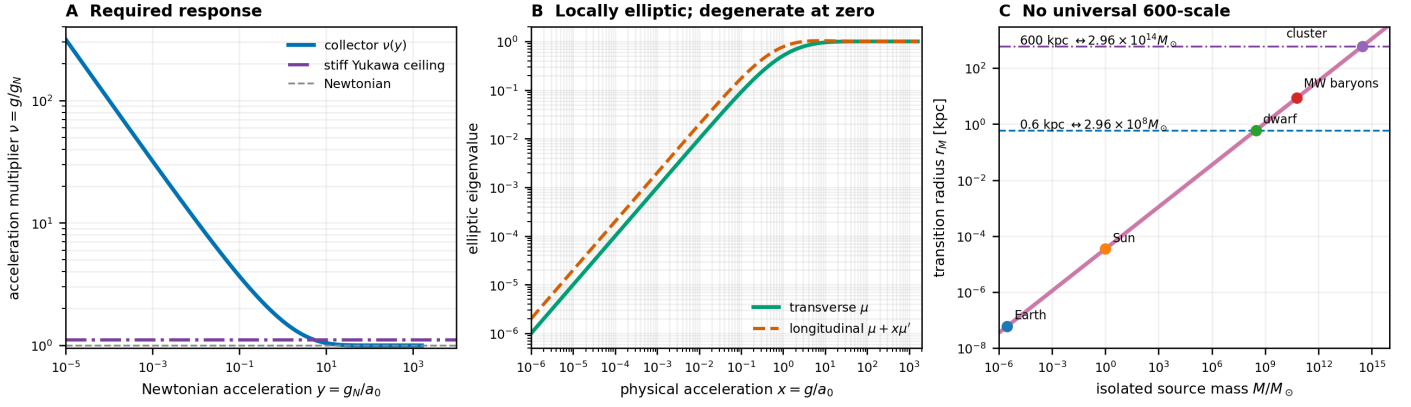


Figure 5: Nonlinear action target reconstructed from the train-frozen collector. Left: the required acceleration response exceeds the rigid positive-Yukawa ceiling at low acceleration. Centre: both local elliptic eigenvalues of Eq. (58) are positive on the plotted nonzero-field domain but tend to zero in the deep limit. Right: the same acceleration threshold maps to different radii for different source masses; neither 0.6 nor 600 kpc is a universal switch. The construction is phenomenological and nonrelativistic, not a derivation from the current HOLO bulk action.

five-dimensional action:

$$\delta(\sqrt{-G}G^{AB}) = \sqrt{-G} \left(\frac{h}{2} \bar{G}^{AB} - h^{AB} \right). \quad (66)$$

It multiplies $\partial_A \delta \chi \partial_B \delta \chi$ with coefficient $-1/(4\kappa_5^2)$. Therefore the fixed-metric canonical scalar sector has no derivative cubic at all; $V''' \delta \chi^3$ is derivative-free. The needed vertex must come from metric-scalar mixing and its constraints. Let π denote the retained light scalar, $Y = -\partial_\mu \pi \partial^\mu \pi / 2$, and σ_a the six heavier modes. A gauge-invariant cubic reduction would have the form

$$\mathcal{L}_4 = P_0(Y, \pi) - \frac{1}{2} \sigma_a (M^2 - \square)_{ab} \sigma_b + c_a \sigma_a Y + \dots \quad (67)$$

Eliminating the gapped modes gives, with $\mathcal{L} \supset -P(Y)$,

$$\Delta \mathcal{L}_{\text{exch}} = \frac{1}{2} Y \mathbf{c}^\top (M^2 - \square)^{-1} \mathbf{c} Y = -\Delta P_{\text{exch}}. \quad (68)$$

Thus a regular finite tower is analytic and keeps the source exponent equal to one; retaining the nonlinear equations does not alter this conclusion while the Hessian remains invertible. Its exchange contribution to P has negative Y^2 coefficient. The total coefficient also contains a direct quartic contact from S_4 , so neither its value nor sign follows from S_3 alone. The local expansion is controlled only for $p\ell \ll \mu_1 = 1.224$.

The required bulk S_3 route can now be made explicit rather than left as an unspecified ADM exercise. The gauge-invariant active-scalar equation of Bianchi et al. (2003) maps to the present conventions through $\phi_{\text{BMP}} = \chi/2$ and $W_{\text{BMP}} = W/4$. Its self-adjoint density is

$$\begin{aligned} \rho_{\text{BMP}} &= e^{4A} \left(\frac{W_{\text{BMP}, \phi}}{W_{\text{BMP}}} \right)^2 = 4e^{4A} \frac{\chi'^2}{W^2} = \frac{2}{3} p, \\ \rho_{\text{BMP}} e^{-2A} &= \frac{2}{3} w. \end{aligned} \quad (69)$$

Both identities hold pointwise on all 1,979 background samples with maximum relative error 7.75×10^{-16} ; the seven eigenvalues are unchanged to 2.15×10^{-12} . Thus this is exactly the already certified local scalar operator, not a new phenomenological ansatz. In comoving gauge its invariant obeys $\tilde{a} = h_{\text{BMP}}/4$. This h_{BMP}

multiplies the transverse scalar projector with coefficient $1/3$ and is not normalized as the earlier isotropic benchmark trace h_R : for that representative $h_{\text{BMP}} = 3h_R/4$ and $\tilde{a} = 3h_R/16$. The constant leaves the operator unchanged but cannot be dropped from an absolute coupling. More importantly for the stabilized branch, the linear Boos scalar g maps, up to its common canonical factor, as

$$\tilde{a}_n = e^{-2A} \left[-g_n + \frac{W}{\chi'^2} g'_n \right]. \quad (70)$$

The transformed seven stiff profiles satisfy the Bianchi-Mück-Prisco bulk equation with maximum rms relative residual 2.43×10^{-4} and peak 1.80×10^{-3} . Their explicitly Bose-symmetric holographic three-point kernel therefore supplies an independent bulk EOM/correlator oracle on the correct linear profiles; it is not by itself the compact S_3 action.

This closes the previous bulk-variable obstruction but not the physical coefficients c_a . Equation (70) must be combined with the finite-endpoint cubic junction/GHY terms and the boundary pieces induced by the field redefinition used in the symmetric kernel. Those terms are microscopic data: adding $\eta_i (\chi - \chi_i)^3 / 6$ to a brane potential changes neither the background nor its certified S_2 spectrum but changes finite-stiffness S_3 . Consequently a value of η_i or a controlled nonlinear stiff limit must be frozen before projecting c_a . The direct S_4 contact remains a separate derivation.

The physical-action route has now passed its required quadratic backward test. In unitary radial gauge, define $q = A + \zeta$, $B_{\mu\nu} = D_\mu D_\nu \beta$, $B = B^\mu{}_\mu$, and $N_\mu = \partial_\mu \beta$. The Einstein-dilaton action including the standard GHY cancellation of radial second derivatives has the exact local density

$$\begin{aligned} \mathcal{L}_{\text{ADM}} &= e^{4q} \left\{ N[R[q] - V] + \frac{F}{N} \right\}, \\ F &= 12q'^2 - 6q'B + B^2 - B_{\mu\nu} B^{\mu\nu} - \frac{1}{2} \tilde{\chi}'^2, \\ R[q] &= -6e^{-2q} [\square \zeta + (\partial \zeta)^2]. \end{aligned} \quad (71)$$

Variation of N gives $N^2 = F/(R - V)$ on the continuous positive-lapse branch. At first order the constraints give $\alpha_1 = \zeta'/A'$ and $\square \beta_1 = \square \zeta/A' + e^{2A} \epsilon \zeta'$. Substitution and integration by parts

recover

$$S_2 = -\frac{3}{2\kappa_5^2} \int du d^4x \left[p \zeta'^2 + w \partial_\mu \zeta \partial^\mu \zeta \right], \quad (72)$$

$$p = e^{4A} \epsilon, \quad w = e^{2A} \epsilon.$$

Thus its mode equation is exactly $-(pf')' = m^2 w f$. Nine independent compact radial/periodic probes reproduce Eq. (72) directly from Eq. (71) with maximum relative discrepancy 3.44×10^{-11} . This closes the same-variable bulk S_2 gate on compactly supported probes before any cubic coefficient is evaluated; the finite-endpoint Neumann action and matter residue remain separate gates.

The finite-boundary bookkeeping is also sharper. In the interval/ Z_2 convention used here,

$$S_{\partial M} = \frac{1}{\kappa_5^2} \sum_i \int_{\Sigma_i} \sqrt{-\hat{\gamma}} \hat{K} - \frac{1}{2\kappa_5^2} \sum_i \int_{\Sigma_i} \sqrt{-\hat{\gamma}} \lambda_i(\hat{\chi}), \quad (73)$$

which yields $n \cdot \partial \chi + \lambda'_i = 0$ and $K_{\mu\nu} - K \gamma_{\mu\nu} + \lambda_i \gamma_{\mu\nu}/2 = 0$. These background junctions fix only $\lambda_i(\chi_i)$ and $\lambda_{i,\chi}(\chi_i)$; they do not fix a functional brane action. There is, however, one decisive conditional branch. With outward orientations $s_- = -1$, $s_+ = +1$, declare

$$A' = -\frac{W}{6}, \quad \chi' = W_{,\chi}, \quad (74)$$

$$V = \frac{1}{2} W_{,\chi}^2 - \frac{1}{3} W^2, \quad \lambda_i(\chi) = -s_i W(\chi).$$

as functional identities along the entire monotone flow. Both junctions then hold at every radial position, so the endpoints can slide. The gauge-invariant collective is the proper interbrane distance

$$R(x) = \int_{Y_-(x)}^{Y_+(x)} N(x, u) du, \quad qquad \delta R = \xi_+ - \xi_- + \int_{u_-}^{u_+} \alpha du. \quad (75)$$

It is not obtained by simply relabelling the lowest finite- γ KK eigenvector; bending must be retained until Eq. (75) is formed. In fact R is not the only physical scalar. Transforming the two exact $m^2 = 0$ solutions to the finite endpoint coordinates $y^\alpha = (Y_-, Y_+)$ gives the Planck-normalized Einstein-frame metric

$$\hat{K}_{ab} = \frac{3}{2} \begin{pmatrix} h_-(h_- + 2A'_-) & -h_- h_+ \\ -h_- h_+ & h_+(h_+ - 2A'_+) \end{pmatrix}, \quad h_i = \frac{e^{2A_i}}{F}. \quad (76)$$

Its eigenvalues are $(0.125269, 2.00317) > 0$: the endpoint map is invertible and there are two physical moduli, not one radion plus a gauge zero mode. This agrees with the independent functional-BPS moduli-space reduction of [Palma & Davis \(2004\)](#) after transforming their pre-Weyl equations to endpoint coordinates. A canonical tangent can therefore be called q only after a physical mechanism fixes its orthogonal modulus.

The static ADM reduction now closes exactly. For $\Delta = [e^{4A} W]_-^+$, the unsimplified Einstein-scalar action, GHY term and the two functional branes give

$$S_{\text{bulk}} = \frac{\Delta}{6\kappa_5^2}, \quad S_{\text{GHY}} = -\frac{2\Delta}{3\kappa_5^2}, \quad (77)$$

$$S_{\text{brane}} = \frac{\Delta}{2\kappa_5^2}, \quad S_{\text{on-shell}} = 0.$$

This is the Hamilton-Jacobi/fake-BPS identity, not an assumed numerical cancellation. Direct integration of the raw reconstructed density on five subintervals of the actual 1,979-point

background reproduces it to a maximum relative residual 1.51×10^{-8} ; flipping one brane sign fails the same test. Hence the potential is flat throughout the allowed BPS endpoint-moduli patch. Along every regular canonical direction in that patch,

$$m^2 = V''_{\text{eff}} = 0, \quad u_4 = V''''_{\text{eff}} = 0, \quad V^{(6)}_{\text{eff}} = 0. \quad (78)$$

Thus the BPS branch proves the two requested zeros, but it also kills the positive sextic required by the tricritical selector. This agrees with the known unstabilized radion of two BPS branes ([Cynolter, 2002](#)).

The earlier finite- γ_i spectrum does not contradict this result: adding $\gamma_i(\chi - \chi_i)^2/2$ changes the brane theory and pins the endpoint. Its Lesgourgues-Sorbo/Boos operator, two finite-element representations and target-free shooting have been cross-checked on the real background, but the direct one-interval EH+GHY+bending derivation and absolute normalization remain open. The synthetic inclined-brane fixture checks coordinate covariance only and is not physical evidence. A full S_3 - S_6 projection is meaningful only after a functional brane theory is selected. For the minimal BPS branch it must reproduce Eq. (78); a nonzero positive q^6 or $q^2 Y$ would require additional microscopic structure.

After the orthogonal moduli have been fixed, the invariant pullback $X_i = \delta \chi_i + \chi'_i \xi_i = Z_i q + O(q^2)$ makes a cubic brane jet begin at q^3 and generically induces quartic massive-mode exchange; it is not a sextic. A clean conditional candidate is instead

$$\lambda_i = -s_i W + \frac{\rho_i}{6!} X_i^6, \quad (79)$$

$$V_E(q) = g_6 q^6 + O(q^7), \quad g_6 = \frac{1}{1440\kappa_5^2} \sum_i \mathcal{W}_{i0} \rho_i Z_i^6,$$

where $\mathcal{W}_{i0} > 0$ is the warp/Weyl weight. Positive ρ_i gives $g_6 > 0$; the first five jets stay BPS and heavy-mode feedback starts at q^{10} . However, ρ_i is new brane physics, not a bulk prediction.

This potential does not generate $q^2 Y$. The induced matter metric supplies an earlier gate. Let

$$F(Y_-, Y_+) = \int_{Y_-}^{Y_+} e^{2A} du, \quad g_E = \frac{F}{F_0} g_4, \quad (80)$$

$$\mathcal{A}_i^2 = e^{2A(Y_i)} \frac{F_0}{F}.$$

Then brane matter is $S_{m,i}[\mathcal{A}_i^2 g_E, \Psi_i]$ and the normalized Jordan curvature selector is $C_i = \mathcal{A}_{i0}^2 / \mathcal{A}_i^2$. On the explicit separation slice $Y_- = u_-$, $Y_+ = R$,

$$C'_-(R_0) = \frac{e^{2A_+}}{F_0} = 0.0197499, \quad (81)$$

$$C'_+(R_0) = \frac{e^{2A_+}}{F_0} - 2A'_+ = 4.249949.$$

Both are nonzero on the real 1,979-point background; indeed they stay positive along the whole monotone slice. Equivalently, $\delta S_{m,i} = \int \sqrt{-\gamma_i} \alpha_i T_i \delta R$ has $\alpha_- = -0.00987497$ and $\alpha_+ = -2.12497450$ per dimensionless radial unit. Thus minimally induced endpoint matter has an odd response before a pure $q^2 Y$ vertex can lead. Dropping F_0/F would erase the lower-brane term, but would also leave a field-dependent Planck coefficient and is not an Einstein-frame calculation.

This slice no-go is not a no-go for every direction. Using Eq. (76) and its Levi-Civita connection, either brane separately

has a unit tangent e_q^a satisfying $e_q^a C_{i,a} = 0$ and

$$e_q^a e_q^b \nabla_a \nabla_b C_i = \begin{cases} -0.330977, & i = -, \\ -0.257821, & i = +. \end{cases} \quad (82)$$

Thus a nonzero even selector jet exists algebraically on the real geometry. It is not yet a physical $q^2 Y$ vertex: the two gradients have determinant -4.84465 , so no tangent is silent for matter on both branes; neither the BPS action nor positive local diagonal quadratic or sextic brane terms select one of the individual silent directions; and a minimally coupled matter scalar weights its Einstein-frame kinetic operator by $\mathcal{A}_i^2 \propto 1/C_i$, not C_i . Separating its standard $-Y$ term leaves the correctly signed candidates $-0.165489 q^2 Y$ and $-0.128911 q^2 Y$ in the displayed Planck-normalized coordinate. However, the constitutive Y has not been identified with that matter operator, and the tangent and absolute normalization remain unselected. Choosing $e_q \perp C_i$ after seeing the target would be inverse design. The sextic detuning can stabilize a chosen direction but cannot select it or repair this matter gate.

The nearest target-independent selector tested is a fixed warped volume F . Its unit tangent is only 0.0686° from the lower-brane silent direction, but leaves $C_{-,a} e_F^a = 9.69 \times 10^{-4}$; exact alignment is equivalent to $A'_- = 0$ (or $W_- = 0$), outside the certified interval. More decisively, on the actual $F = \text{constant}$ curve that formal limit gives $C_- = 1 + 0.00227481 q^2 + \dots$, so minimal $-Y/C_-$ has the wrong positive $q^2 Y$ sign. Defining $s = C_- - 1$ would reverse it, but requires an additional nonminimal operator; no global- F constraint is present in the action.

An exact classical half-power mechanism is instead

$$s = q^2 \geq 0, \quad \mathcal{L}_{\text{aux}} = -sY + m_2 s + \frac{u_4}{2} s^2 + \frac{1}{3} s^3. \quad (83)$$

At the tricritical point $m_2 = u_4 = 0$, stationarity gives $s = \sqrt{Y}$ and $P(Y) = sY - s^3/3 = 2Y^{3/2}/3$ exactly. Any positive m_2 introduces a threshold, and any positive u_4 restores $P \propto Y^2$ in the deep limit. Moreover, the stationary curvature is $8Y$, so a propagating q becomes gapless as $Y \rightarrow 0$ and cannot be integrated out uniformly as a local auxiliary variable. The alternative identity

$$\frac{2}{3} Y^{3/2} = \frac{4}{3\pi} \int_0^\infty dm \frac{Y^2}{Y + m^2} \quad (84)$$

requires a constant gapless density per mass. The present seven modes pass through logarithmic slope $3/2$ for only 0.210 dex and are not such a continuum. Equation (84) is also an identity, not a healthy generation proof: ordinary stable Gaussian elimination has the opposite stationary sign and generally yields a nonlocal momentum kernel.

The simplest Lorentz-covariant use of $P \propto Y^{3/2}$ does not yet close those gates. On a spacelike background its longitudinal-to-transverse principal coefficient ratio is $(P_Y + 2Y P_{YY})/P_Y = 2$, while both coefficients vanish at $Y = 0$. Furthermore, at linear order a purely conformal matter metric gives $\Phi_J = \Phi_E + \alpha \delta \phi$ and $\Psi_J = \Psi_E - \alpha \delta \phi$, so the direct scalar contribution cancels from $\Phi_J + \Psi_J$. A scalar acceleration is therefore not by itself a lensing prediction.

The next test is now pre-registered rather than inferred from the target. After freezing S_5 , all boundary actions, κ_5 , ℓ , A_m , matter localization, and two conserved source geometries, one must solve the nonlinear bulk and extract its Dirichlet-to-Neumann

flux $\Pi(g)$. Acceptance requires

$$\lim_{g \rightarrow 0} \frac{\Pi(g)}{g} = 0, \quad \lim_{g \rightarrow 0} \frac{\Pi(g)}{g^2} = \frac{1}{a_{0,\text{bulk}}} > 0, \quad (85)$$

$$\lim_{g \rightarrow \infty} \frac{\Pi(g)}{g} = 1.$$

together with mass slope $1/2$, geometry-independent local constitutive response, a stable causal fluctuation operator, and a separately computed lensing potential. The current checkout cannot yet run this physical test because the boundary completion, absolute scale, matter localization, and Wilson coefficients remain unfixed. Choosing any of them after revealing the galaxy target would be inverse design.

For provenance, the five-parameter capped legacy P5 dictionary was also refitted on the physical training inputs under its frozen bounds. It reaches all five bounds and gives $\chi^2/\text{point} = 290.98$ on test. It is rejected as a historical comparator and is not used as the current HOLO curve. A nested generalized RAR with $v_q(y) = [1 - \exp(-y^q)]^{-1/(2q)}$ returns $q = 0.497$; its 0.13 per cent test gain is immaterial and the extra parameter is not selected. The result is an empirical target, not a detection and not an action-derived HOLO galaxy force law. The present action-derived stiff force is available and falsifiable, but is empirically insufficient; the additional sector required by Eqs. (55)–(56) has not been derived.

9 Dictionary-dependent cosmological growth

Under Eq. (46), a scalar-derived equation-of-state template is inserted into a matter-plus-dark-energy background. The standard linear growth equation is

$$D_{,NN} + \left(2 + \frac{d \ln H}{dN}\right) D_{,N} - \frac{3}{2} \Omega_m(N) D = 0, \quad (86)$$

with $f\sigma_8 = \sigma_{8,0}(d \ln D / d \ln a)D$. At the three BOSS DR12 redshifts, the stored covariance calculation gives

$$\chi_{\text{dict}}^2 = 2.266, \quad \chi_{\Lambda\text{CDM}}^2 = 2.443. \quad (87)$$

Thus $\Delta\chi^2 = -0.177$, a statistically negligible difference at current precision. The stored high-redshift suppression, approximately -12 per cent near $z_{\text{cos}} = 1$, is a falsifiable prediction of the *composite dictionary model*, not of the five-dimensional bulk action alone.

10 Prediction-factory diagnostics

The revised calculation is organized as a fail-closed graph of physical links: an output is promoted only when the action, boundary data, dimensional scale, and evaluation protocol required by the corresponding arrow are present. Four new checks illustrate the distinction between generating a prediction and retrofitting a result.

Boundary branches. No observational input enters this step: all four separated self-adjoint endpoint choices use the same bulk coefficients. The NN branch has its exact zero mode. Replacing only the IR condition gives an ND lightest mode

$$\mu_{\text{ND},0} = 0.00274498, \quad \beta_{\text{UV},0} = 0.05429095. \quad (88)$$

Table 4: What the previous sector supplied versus the critical constitutive architecture. “Exact” refers to the exponent algebra, not to a physical bulk derivation.

Test	Previous fixed-pole sector	Critical constitutive sector
Carrier	Seven positive, fixed, gapped canonical modes	Nonlinear collective amplitude or a gapless continuum
Deep source scaling	$g \propto M$	$g \propto \sqrt{M}$ exactly
Small-field action	Analytic $Y + O(Y^2)$	Nonanalytic $2Y^{3/2}/3$ at the selected critical point
What is derived	Quadratic carrier, spectrum, and linear residues	Mathematical selector and decisive falsifiers
What remains	Cubic overlaps c_a and direct S_4 contact	q^2Y , $m_2 = u_4 = 0$, positive sextic normalization, a_0 , causal completion, sourcing, slip, and lensing

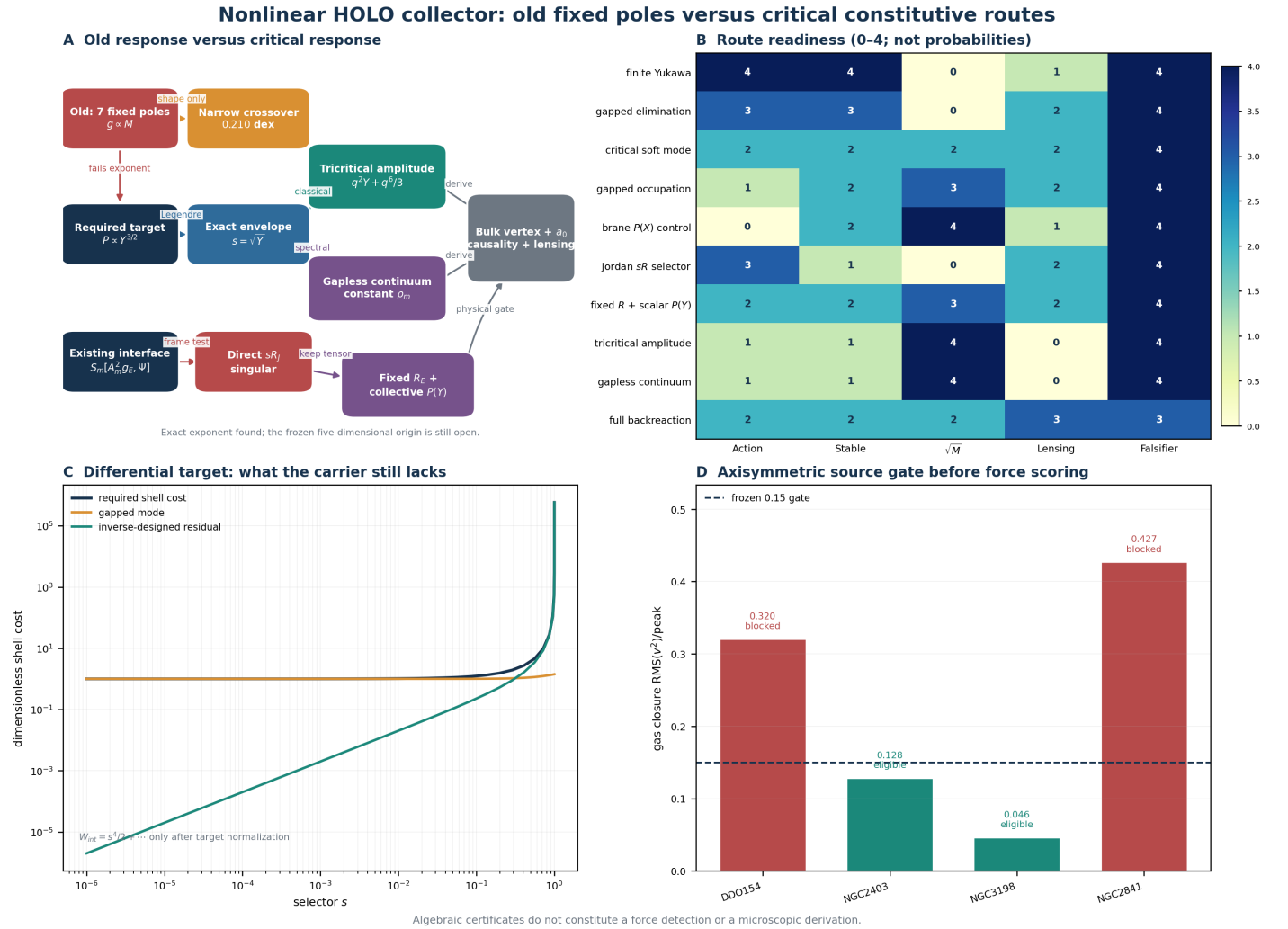


Figure 6: Generated nonlinear routes and their present gates. The exact Legendre envelope specifies the required selector cost, not its origin. The old seven-pole response has the wrong mass exponent and crosses the desired slope for only 0.210 dex. A tricritical amplitude and a constant gapless spectral density reproduce the exponent exactly, but neither has yet been generated healthily by the frozen bulk action. The exact Jordan-frame map diagnoses why using s as the full coefficient of R_j is singular as $s \rightarrow 0$. The surviving architecture keeps the tensor Einstein–Hilbert term fixed and seeks a microscopic derivative scalar or collective sector. The lower-right source gate admits only two of the four bounded axisymmetric prototypes. Scores are readiness diagnostics, not probabilities or evidence for a new force.

SPARC audit: action-derived stiff force versus empirical target

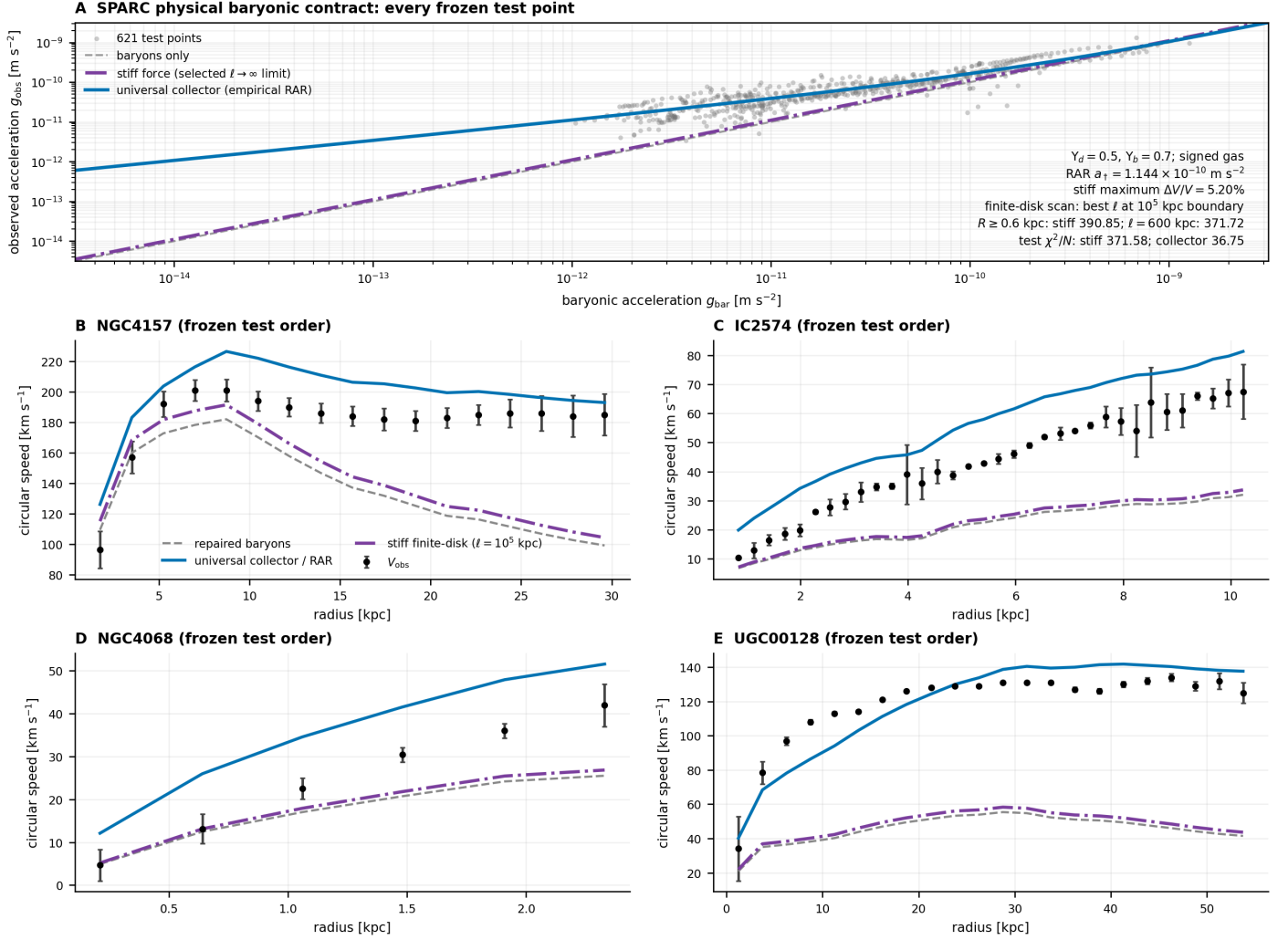


Figure 7: Physical-input SPARC audit. The upper panel shows every point in the frozen test split against the train-frozen universal collector (the empirical RAR target) and the stiff long-range limit selected by the finite-disk scan. The lower panels show the first four test IDs in the predetermined split order, not selected for visual agreement. Blue is the empirical collector target; purple is the geometry-matched stiff candidate. The visible misses are retained rather than disguised by fitting each test galaxy.

The latter is 0.0542910 at seven decimal places. An independent continuous-shooting calculation gives $\mu_{\text{ND},0} = 0.0027447613$, a relative difference 7.84×10^{-5} . The former zero mode is therefore ultralight rather than removed. DN and DD start at $\mu = 0.91382$ and 1.23053 , respectively, but an exact UV point probe obeys Dirichlet data and decouples. No branch is selected: that requires a boundary or junction action fixed before testing.

The minimal positive Robin boundary-potential family makes one candidate action explicit, with no boundary kinetic terms:

$$Q[f] = \int du p(f')^2 + b_{\text{UV}} f_{\text{UV}}^2 + b_{\text{IR}} f_{\text{IR}}^2, \quad b_{\text{UV}}, b_{\text{IR}} \geq 0. \quad (89)$$

It implies $-p f'_{\text{UV}} + b_{\text{UV}} f_{\text{UV}} = 0$ and $p f'_{\text{IR}} + b_{\text{IR}} f_{\text{IR}} = 0$. The dimensionless scan coordinate is $\rho_i = b_i/C_p$, where $C_p = (\int du/p)^{-1} = 6.8836329 \times 10^{-4}$. A prospective two-parameter scan finds that IR stiffness alone cannot raise the lightest mode above $\mu_0 = 0.002744976$. UV stiffness raises that mass continuously. At much larger stiffness the first pair undergoes an avoided crossing and exchanges UV residue near $\rho_{\text{UV}} \approx 1.12 \times 10^5$; the minimum sampled first-pair gap is 0.0119229. The independently finite-differenced identity

$$\frac{\partial \mu_n^2}{\partial b_{\text{UV}}} = f_n(u_{\text{UV}})^2 = \frac{3\beta_n^2}{I_g} \quad (90)$$

is the Hellmann–Feynman link between boundary stiffness, pole motion, and material residue. The map makes the consequences of any prospectively selected boundary coefficients falsifiable; it does not select them.

Conditional functional superpotential branes. For the reconstructed canonical scalar χ , the background fixes $W(\chi(u)) = -6A_{,u}$ and $\chi_{,u} = W_{,\chi}$. It fixes only the value and first derivative of each endpoint potential. If one additionally declares the functional ansatz $\lambda_- = W$ and $\lambda_+ = -W$, the gauge-invariant scalar boundary coefficients then satisfy

$$\frac{g_-}{a} = -\frac{\chi_{,uu}}{\chi_{,u}} + \lambda_{-, \chi\chi} = 0, \quad \frac{g_+}{a} = -\frac{\chi_{,uu}}{\chi_{,u}} - \lambda_{+, \chi\chi} = 0. \quad (91)$$

The two-brane zero-mode theorem then requires at least one massless gauge-invariant scalar (Lesgourgues & Sorbo, 2004); the exact action calculation above identifies the relative radion and proves its flat potential. This is a conditional brane ansatz, not a brane action uniquely derived from the bulk. Adding local positive curvatures,

$$\lambda_- = W + \frac{\gamma_-}{2}(\chi - \chi_-)^2, \quad \lambda_+ = -W + \frac{\gamma_+}{2}(\chi - \chi_+)^2, \quad (92)$$

gives $g_-/a = \gamma_- > 0$ and $g_+/a = -\gamma_+ < 0$. Their values are new boundary-sector inputs; they are not determined by the bulk solution. The stiff limit $\gamma_{\pm} \rightarrow \infty$ is therefore declared as one parameter-free limiting candidate rather than inferred from SPARC. A gauge-invariant finite-element solve gives

$$(\mu_0, \dots, \mu_3) = (0.290502, 1.224184, 1.725453, 2.188327), \\ (\mu_4, \dots, \mu_6) = (2.744438, 3.344908, 3.961150), \quad (93)$$

while an independent DOP853 shooting calculation agrees to a maximum relative mass difference 7.36×10^{-5} . This cross-checks the translated linear operator. The absolute one-interval normalization and physical matter residue still require a direct EH+GHY+bending reduction; Eq. (37) is therefore conditional on that unfinished interface step.

SPARC split result. On the 621 fixed test points, the physical-input scores are 371.58 for the action-derived stiff force, 414.20 for the old P6 trace limit, 414.23 for baryons alone, 290.98 for legacy P5, and 36.75 for RAR. The RAR test median absolute velocity error is 14.5 per cent. The stiff candidate replaces P6 and P5 as the current physical curve, but its fixed residues remain far too small and have the wrong acceleration dependence. Its one-scale finite-disk scan selects only the upper boundary, not a finite ℓ . The legacy P5 optimizer exhausts its budget with all five parameters on bounds, so that trace-backed dictionary is retained only as rejected provenance. Because the catalogue has been used during development, this is a retrospective audit, not a blind confirmation.

Published DESI DR1 entries. Four published ShapeFit-plus-BAO marginal entries inside the frozen trace domain, at $z_{\text{eff}} = \{0.30, 0.51, 0.71, 0.92\}$, were evaluated without refitting (DESI Collaboration et al., 2025). Comparing their quoted $f\sigma_{s8}$ marginals directly with the frozen $f\sigma_8$ curves gives

$$\chi_{\text{dict,diag}}^2 = 2.6917, \quad \chi_{\Lambda\text{CDM,diag}}^2 = 2.4189, \\ \Delta\chi^2 = +0.2729. \quad (94)$$

Both curves are compatible with this low-dimensional diagnostic and the small positive difference does not favour the dictionary curve. Because distances, shape-slope information, and within-vector and cross-bin correlations are not modeled, these numbers are explicitly not the official DESI likelihood.

Wilson route. The new analyser validates SU(3) link matrices and implements rectangular $W(R, T)$, effective potentials, Creutz ratios, blocking, and jackknife uncertainties. It fails closed on the current inputs: the available lattice files contain plaquette/correlator summaries, not link configurations. Consequently neither $a^2\sigma$ nor a continuum string tension is reported.

11 Single-arm response-operator spectroscopy

Let $\{\psi_n\}$ denote the stored response-operator basis and let δa be a vector of modal amplitudes. Each observational arm α has a versioned response operator

$$J_{in}^{(\alpha)} = \frac{\partial y_i^{(\alpha)}}{\partial a_n}, \quad \delta y^{(\alpha)} \simeq J^{(\alpha)} \delta a. \quad (95)$$

The DESI example reconstructs a bounded δG from one residual vector. This is an operational single-arm inverse problem, not a simultaneous multi-arm detection. This stored basis is not identified with the independently normalized trace-carrier modes f_n of Section 5.

For diagnostic comparison, normalized arm blocks are stacked and decomposed as

$$\hat{J}_{\text{multi}} = U\Sigma V^T, \quad (96)$$

after Frobenius-per-output scaling. The singular values summarize shared sensitivity directions; they are not a global reconstruction.

A modeled time dependence may be written

$$\dot{a}_n = \sum_m M_{nm} a_m + S_n, \quad (97)$$

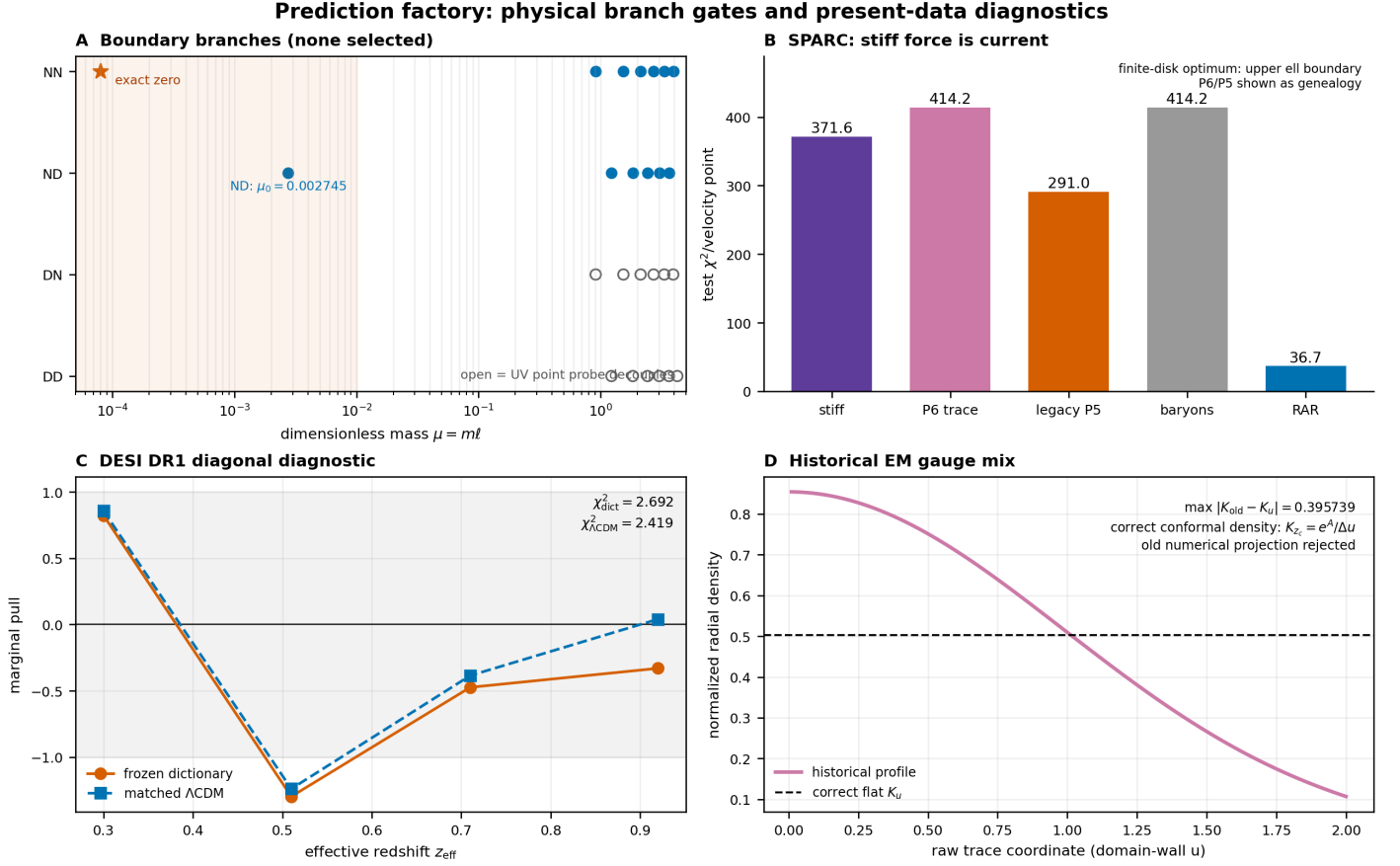


Figure 8: Prediction-factory diagnostics. (A) The four endpoint branches: filled markers couple to the declared UV point probe, while open markers obey UV Dirichlet data; no branch is selected. (B) With the physical SPARC baryonic input, the action-derived stiff force improves on baryons but remains far from the train-fitted empirical RAR; P6 and legacy P5 are shown only as genealogy. (C) Published DESI DR1 marginal pulls give no preference over matched Λ CDM. (D) The historical electromagnetic profile is compared with the coordinate-correct $Z = 1$ domain-wall zero-mode density; their disagreement exposes the old radial-gauge substitution.

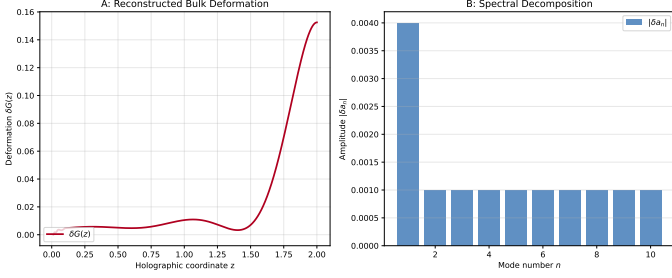


Figure 9: Bounded single-arm reconstruction and modal amplitudes. The plotted deformation is response-operator dependent.

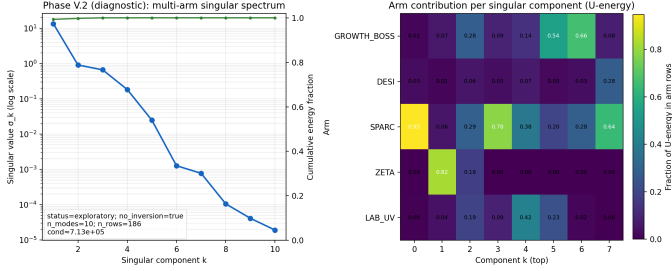


Figure 10: SVD of the stacked normalized response blocks. Diagnostic only; no cross-arm closure is inferred.

where both M and the closure source S must be recorded. Observable projections such as $\Delta f = \sum_n \kappa_n a_n$ remain conditional on their declared coupling ansatz.

12 Dictionary-based geometric clock and legacy laboratory interface

The five-dimensional domain-wall curvature is

$$R_{5,\text{phys}} = \frac{\widehat{R}_5}{\ell^2}, \quad \widehat{R}_5 = -8A'' - 20A'^2 = \frac{1}{2}\chi'^2 + \frac{5}{3}V. \quad (98)$$

A fresh audit verifies the second, independent Einstein-trace identity to 2.84×10^{-14} and reproduces the same scalar pointwise after transforming to conformal radial coordinates. It also finds that the stored clock evaluated curvature by treating the solver field $(dA)_{\text{stored}}$ as A' , although the reproduced trace satisfies $A' = u(dA)_{\text{stored}}$. On their common certified slice the stored curvature spans -21.60 to -19.09 , while the corrected value spans -13.25 to -73.47 ; their rms difference is 26.91. The old plot below is retained solely as provenance.

A corrected geometry-only protocol factor can instead be declared as

$$g_R(u) = \frac{e^{A(u)} \sqrt{|\widehat{R}_5(u)|}}{e^{A_0} \sqrt{|\widehat{R}_{5,0}|}}, \quad \Theta_R(u) = \int_1^u g_R(s) ds. \quad (99)$$

Equation (99) is a deterministic dimensionless curvature phase, not proper time and not a mode amplitude. On a physical compact branch it would define $\omega_R = c\sqrt{|\widehat{R}_5|}/\ell$, while $\omega_n = c\mu_n/\ell$; hence ω_n/ω_R is geometry-fixed but seconds still require ℓ . The historical LOCK5 instead inferred ξ from the two frequencies it subsequently “verified” and mixed this bulk curvature with $R_4 \simeq 8\pi G\rho/c^2$. Its location table is therefore a circular calibration, not a Ricci-clock prediction.

The historical UV-screened laboratory channel was tabulated on the raw trace grid, labelled that grid as conformal z , and used

$$K_{\text{old}}(z) = \frac{e^{A(z)}}{\int e^{A(s)} ds}, \quad y_{\text{old}}(t) = \int_{z_{\text{UV}}}^{z(t)} K_{\text{old}}(s) \delta(s) ds. \quad (100)$$

The effective-action audit identifies that raw grid as domain-wall u . Constructing the conformal coordinate instead requires $dz_c = e^{-A(u)} du$ and gives a conformal span 4.90458, not the raw-grid span. For a flat photon with $Z = 1$, the coordinate-correct domain-wall density is constant, $K_u = 0.507422$, whereas the historical profile differs from the corresponding full-grid constant by as much as 0.395739. Thus the previous near-machine-precision “recovery” was an identity produced by setting $A = \log K_{\text{old}}$, not a check against the reconstructed geometry.

For the stored Yb comparison ($n = 595$), the historical projection gives $\chi^2/n = 22.59$ and Pearson $r = -0.0554$. This null/poor-fit historical comparison remains useful negative provenance, but it is not a test of the coordinate-correct response and does not constitute a clock detection.

A consistent completion starts from the bulk action

$$S_\gamma = -\frac{1}{4g_5^2} \int d^5x \sqrt{-g} Z(\chi) F_{MN} F^{MN}. \quad (101)$$

For $ds^2 = e^{2A(u)} \eta_{\mu\nu} dx^\mu dx^\nu + \ell^2 du^2$, the dimensionless photon eigenvalues and profiles obey

$$-\partial_u (e^{2A} Z \partial_u f_{\gamma,n}) = \mu_{\gamma,n}^2 Z f_{\gamma,n}, \quad \int du Z f_{\gamma,m} f_{\gamma,n} = \delta_{mn}, \quad m_{\gamma,n} = \mu_{\gamma,n}/\ell. \quad (102)$$

In conformal gauge the same equations contain the weight $e^A Z$ and the two normalized densities are

$$K_u = \frac{Z|f_\gamma|^2}{\int du Z|f_\gamma|^2}, \quad K_{z_c} = \frac{e^A Z|f_\gamma|^2}{\int dz_c e^A Z|f_\gamma|^2}, \quad K_u du = K_{z_c} dz_c. \quad (103)$$

The cumulative probability measures agree to 6.27×10^{-12} . Equation (103) is the covariant functional form underlying the old ansatz, but Eq. (100) was not its numerical realization.

The scalar-photon vertex is also fixed at this conditional action level. In the same almost-radial gauge used for the trace carrier, the linear constraint of Arutyunov et al. (2000) gives

$$A_u h_{uu} = \frac{1}{4} \partial_u h, \quad N = \frac{1}{2} h_{uu} = \frac{\partial_u h}{8A_u}. \quad (104)$$

The four-dimensional conformal trace cancels from $\sqrt{-g} F_{\mu\nu} F^{\mu\nu}$, but the radial lapse N does not. For $Z = e^{c_\gamma \chi}$, comoving endpoints, no brane kinetic terms, and $h = \sum_n q_n f_n$, dimensional reduction therefore gives

$$B_F = 1 + \sum_n d_{\gamma n} \frac{\varphi_n}{M_{\text{Pl}}} + \dots, \quad d_{\gamma n}(c_\gamma) = \sqrt{\frac{I_g}{3}} \frac{\int du e^{c_\gamma \chi} f_n' / A_u}{\int du e^{c_\gamma \chi}}. \quad (105)$$

No coefficient is fitted at $c_\gamma = 0$. For the six positive NN modes, $d_{\gamma n} =$

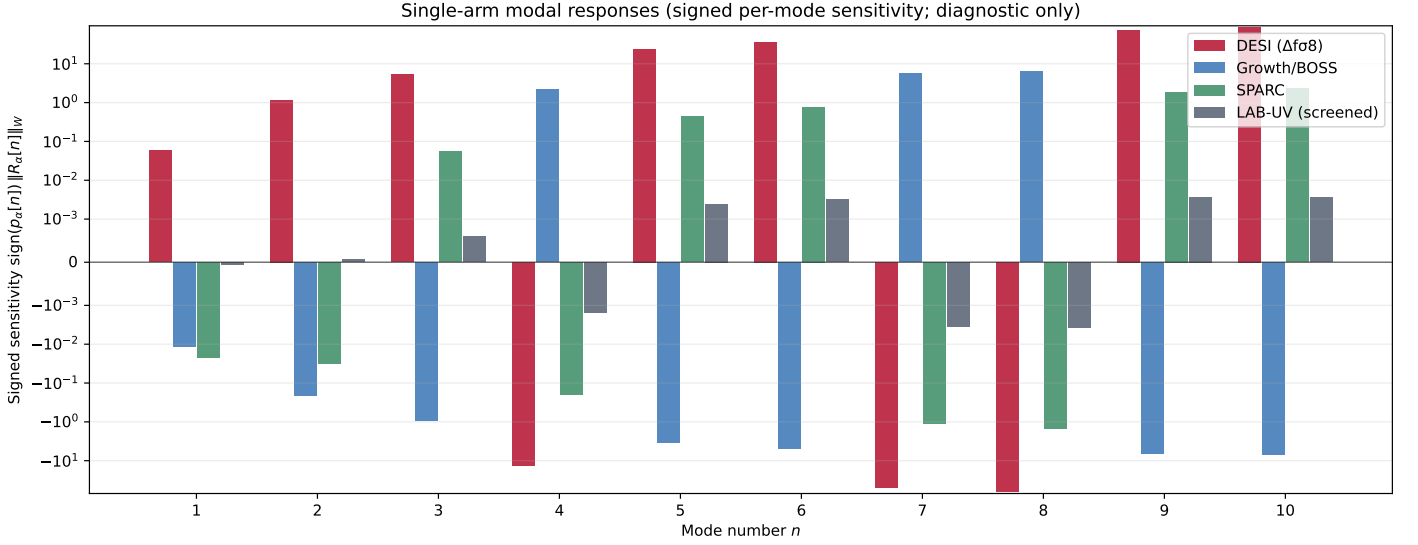


Figure 11: Signed per-mode sensitivities for the separate response arms. The UV-screened laboratory arm remains strongly suppressed.

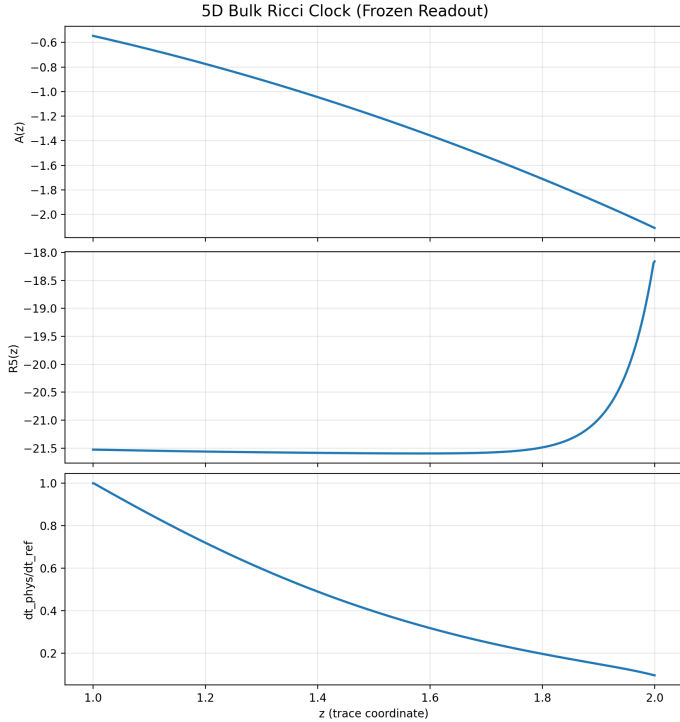


Figure 12: Legacy warp, curvature, and protocol artefact, retained for provenance. Its curvature used the mislabeled stored derivative and is not the corrected $\widehat{R}_5$ of Eq. (99).

(3.94563, -2.52337, 2.90124, -2.42675, 2.83547, -2.44815). Mode signs are conventional, while the source products $\beta_n d_{\gamma n}$ are invariant. The unfitted response to a nonconstant gauge kinetic function is separately fixed by

$$\partial_{c_\gamma} d_{\gamma n}|_0 = \sqrt{\frac{I_g}{3}} \text{Cov}_u \left(\chi, \frac{f'_n}{A_u} \right). \quad (106)$$

Neumann photon data at $c_\gamma = 0$ give the independent dimensionless comb

$$\begin{aligned} \mu_{\gamma,n} &= (0, 0.652597, 1.301427, 1.939255, \dots), \\ \eta_{\gamma,n} &= (1, 0.61936, 0.69848, 0.75808, \dots), \end{aligned} \quad (107)$$

where $\eta_{\gamma,n}$ is the squared UV charge residue relative to the zero mode. With the additional interface assumption of a conserved four-dimensional electric current localized at the UV endpoint, $S_J = q_5 \int d^4x A_\mu(x, u_{UV}) J^\mu$, its minimally coupled static template is

$$V_\gamma(r) = V_{\text{Coulomb}}(r) \left[1 + \sum_{n>0} \eta_{\gamma,n} e^{-\mu_{\gamma,n} r / \ell} \right]. \quad (108)$$

Together with the scalar masses, this is a two-sector or “double-comb” fingerprint tied to the same ℓ . For a point source in the stored sign convention, the corresponding fine-structure response is

$$\frac{\delta \ln \alpha}{U} = \sum_n 2\beta_n d_{\gamma n} e^{-\mu_n r / \ell}, \quad U = \frac{GM}{r}. \quad (109)$$

An atomic comparison would then require $\delta \ln(\nu_A/\nu_B) = \Delta K_\alpha \delta \ln \alpha$. Equations (107)–(109) do not predict hertz until ℓ , a source, atomic sensitivities, and the physical bulk-photon and scalar boundary branches are supplied. Moving endpoints, brane bending, or localized kinetic terms would add boundary contributions and require a new derivation.

The archived Yb-clock and 10-GHz records were produced to demonstrate coherent optical-to-microwave phase transfer: the microwave phase deliberately tracks the optical-clock phase (Nakamura et al., 2020). Their mutual coherence is therefore an engineering observable with a shared reference, not two independent detectors of a universal breathing mode. P7 instead requires

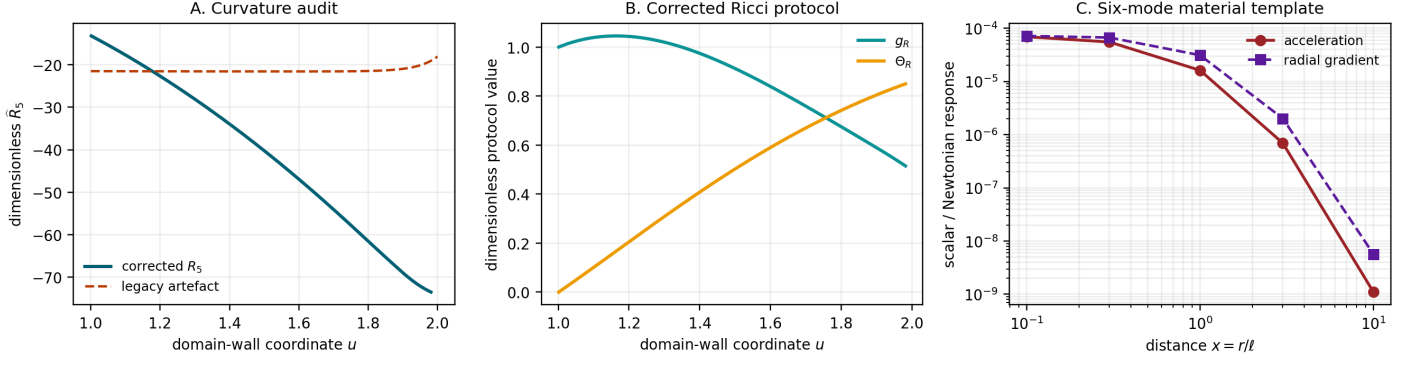


Figure 13: Corrected Ricci/material bridge. Left: corrected curvature from the certified effective action versus the legacy series. Centre: dimensionless curvature cadence and accumulated protocol coordinate. Right: acceleration and tidal templates of the six positive Neumann modes; the physical horizontal scale remains r/ℓ .

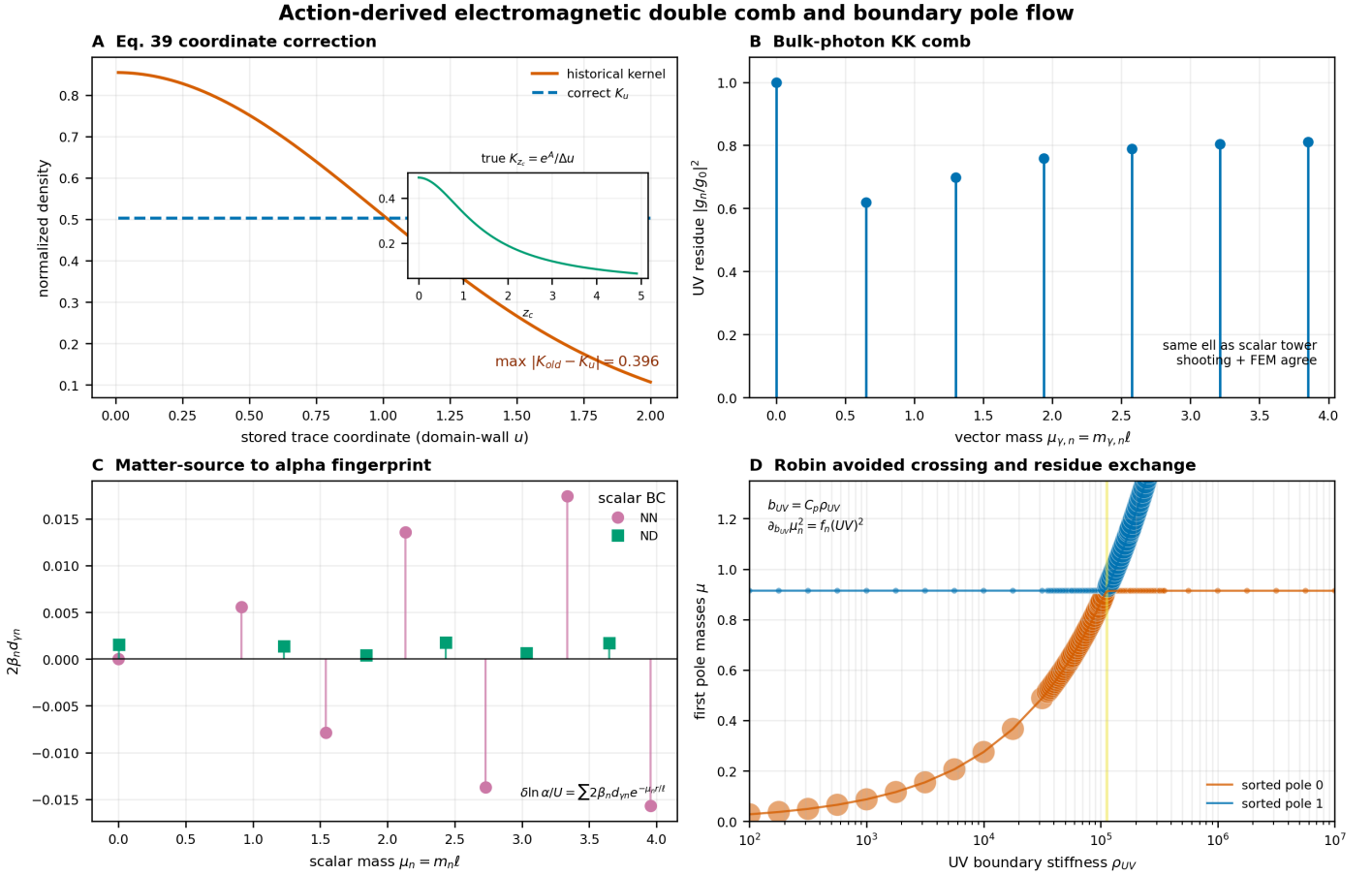


Figure 14: Coordinate-correct electromagnetic completion and double-comb fingerprint. (A) The historical coordinate-mixed profile against the flat bulk-photon density in domain-wall gauge, with the true conformal density inset. (B) Photon Kaluza–Klein masses and UV charge residues. (C) Action-derived source-to- α coefficients for the NN and ND scalar branches. (D) Robin pole motion and UV-residue exchange across the avoided crossing. All axes are dimensionless; no physical boundary coefficient, length scale, source, atomic sensitivity, or detection is inferred.

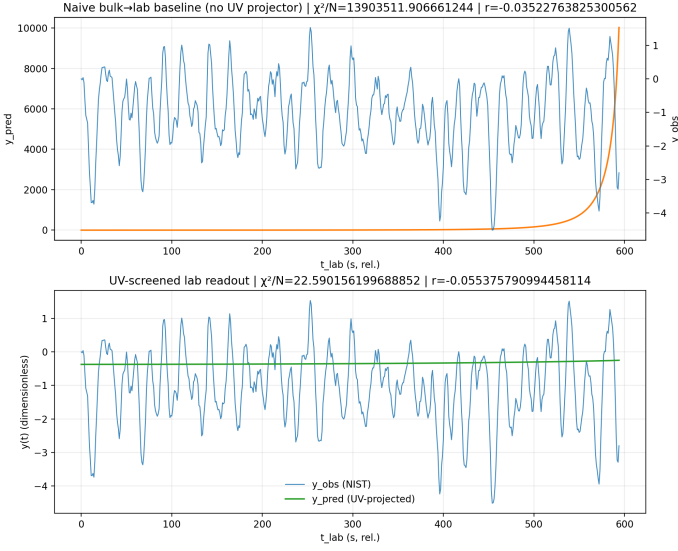


Figure 15: Historical coordinate-mixed clock projections retained for provenance. Their poor correlation is not a detection and is not a test of the corrected bulk-Maxwell response.

separate sites or species, frozen phase and frequency rules, environmental vetoes, and a held-out acquisition.

The Anti-Zeno response table can likewise be composed with the protocol-time dictionary, but it remains an interface hypothesis until its action, units, and independent calibration are specified.

13 Discussion and validation path

The effective reconstruction establishes a nontrivial positive result: the actual frozen geometry admits a ghost-free Einstein-scalar completion on the certified finite interval. It also explains why the deformation passed unchanged into downstream arms despite the preconditioner label error. What it does not do is make $K(\phi)$ and $V(\phi)$ prior predictions; they were inferred from the achieved profiles.

A second positive result is available under a sharper evidence label. The compact-interval probe completion derives a normalized matter coupling without using any observational residual or historical fitted coefficient. Its β_0 and mode tower are consequences of the declared interval and Neumann benchmark, not unique consequences of the bulk geometry alone.

Several previously proposed steps are now executable rather than aspirational. The minimal positive Robin boundary-potential family has been mapped; the electromagnetic radial gauge has been corrected; the lapse constraint supplies a scalar-photon vertex; the bulk photon supplies an independently checked spectral comb; and the Wilson analysis fails closed at the missing link ensemble. The scalar and photon combs share one physical length and can therefore be falsified jointly once that scale and a boundary action are fixed. None of those calculations selects those missing inputs. The remaining forward programme is therefore:

1. compress the tabulated $K(\phi)$ and $V(\phi)$ into a compact analytic form without using observational arms;
2. freeze that action, boundary conditions, and solver tolerances;

3. regenerate A , ϕ , and the canonical χ without reading the old trace;
4. derive the boundary/junction action that fixes a point in the mapped Robin family, and fix ℓ from independent physics;
5. export thermalized SU(3) link ensembles so that the implemented rectangular-loop, static-potential, Creutz, and continuum pipeline can run;
6. recompute the gauge-invariant spectrum with a confining infrared completion and frozen boundary conditions;
7. compare the frozen matter force with fifth-force and post-Newtonian constraints without adding post-hoc screening;
8. drive the derived material transfer law with a calibrated source and test it against a mechanically independent finite-element model and a held-out source distance;
9. test the P7 breathing comb with an independently anchored frequency, source coherence, causal delays, and detectors without a shared reference;
10. derive a four-dimensional cosmological interface before promoting the current radial dictionary to a growth law;
11. choose the bulk- or brane-photon realization prospectively, including photon boundary data, possible localized kinetic terms, source geometry, and atomic response coefficients; and
12. freeze every model and nuisance rule before an independent galaxy, survey-release, or clock-session holdout.

Passing this programme would promote the current operational unification into a forward predictive model.

14 Conclusions

The original work achieved a reproducible shared-background instrument: one frozen warp geometry and deformation feed hadronic, galactic, cosmological, laboratory, and response-operator calculations. The revision identifies that achievement explicitly rather than discarding it.

The previously declared canonical polynomial action and derivative labels do not generate the frozen trace as stated. A geometry-preserving inverse reconstruction supplies the missing bulk completion. Its kinetic function is positive throughout the certified interval, its canonical representation closes the full Einstein-scalar equations at floating-point precision, and it recovers the operational deformation with correlation 0.999999905.

An explicitly conditional compact-interval reduction additionally produces a matter interaction rather than merely naming one: it gives $\beta_n = \sqrt{I_g/3} f_n$, with $\beta_0 = 0.0542901$ and conditional unscreened strength $2\beta_0^2 = 5.89483 \times 10^{-3}$. This result was obtained blind to the observational arms and independently checked numerically. Its physical realization still depends on the boundary/junction completion and on the scale ℓ ; it is neither a detection nor a unique prediction of the bulk trace. Its Neumann massless branch is ruled out in the unscreened universal reading; the retained positive-mode ratios and material transfer equation are therefore a benchmark for a prospectively rederived boundary completion, not permission to delete only the failed zero mode after comparison. The endpoint audit sharpens that

warning: the simplest IR-Dirichlet change leaves a strongly coupled ultralight mode, while UV-Dirichlet branches make the declared point probe vanish. The minimal positive Robin phase map adds a stronger result: positive IR stiffness alone has an ultralight ceiling, while UV stiffness moves both poles and residues through an avoided crossing according to the verified Hellmann–Feynman identity. A boundary action, rather than observational performance, must choose the physical realization.

Restoring time dependence supplies the missing distinction between a static force and a breathing interaction. P7 proves that the canonically normalized stiff force is the zero-frequency limit and predicts the complete correlated set of seven evanescent thresholds, outgoing-wave frequencies, group delays, and relative periods. An independent mode frequency would fix ℓ through Eq. (44). It does not, however, provide the source occupation or boundary action, and the historical 1/600 code-unit frequency cannot serve as that independent clock.

The Ricci and Wilson audits close two further provenance gaps. The corrected $\widehat{R}_5$ supplies a dimensionless curvature cadence but not laboratory seconds, and the old Wilson-labelled number is an Einstein-frame endpoint conversion with external α' , not a world-sheet or lattice area law. Using a QCD scale as a compactification length would require a new UV matching relation and cannot be inferred by chaining the present files.

The electromagnetic audit supplies a constructive bridge and one necessary correction. The functional conformal density $K_{zc} \propto e^A Z(\chi) |f_\gamma|^2$ follows from a bulk Maxwell action, but the historical numerical profile mixed the domain-wall and conformal coordinates. The corrected reduction gives an invariant measure, the action-derived coefficients d_{yn} , and a bulk-photon Kaluza–Klein comb. Its shared ℓ with the scalar tower defines a correlated double-comb target rather than two freely aligned spectra. A brane-localized photon gives a different classical result, and neither branch has yet been selected.

The cosmological diagnostics and UV projection remain reproducible. The junction audit instead finds a flat BPS potential and resolves two physical endpoint moduli with positive kinetic eigenvalues (0.125269, 2.00317). Each endpoint matter selector has an algebraically silent tangent and a nonzero covariant even jet, but no tangent works for both branes and the existing BPS, quadratic and sextic brane terms select none. A physical $q^2 Y$ vertex still requires a derived selector, matter operator and full constraint reduction. A fixed- F candidate nearly aligns the lower selector but is not exact and its true constrained curve gives the wrong minimal-matter sign. The independently checked stiff limit remains a separate candidate. In SPARC its action-derived force improves on baryons, but the geometry-matched finite-range scan reaches the long-range boundary and remains far from the empirical RAR, which is not relabelled as a HOLO force law. P6 and P5 retain only numerical provenance. Equation (56) specifies the missing signed response, while Eqs. (57)–(59) reconstruct a locally elliptic but zero-field-degenerate target with a mass-dependent transition radius. The embedding gate excludes that half-power scaling from the present regular linearized sector, not from every possible nonlinear or derivative completion. Completing the force therefore still requires a derived state-dependent coupling, a physical selection of the moduli tangent, three-dimensional source and boundary data, and independent observation. A single compactification length also cannot represent both the legacy QCD proxy and galactic force: their conditional mismatch is 40.94 orders of magnitude. The apparent galactic clock is a boundary value, DESI gives no preference, and

the Wilson route correctly reports missing input. These distinctions turn the overextended claim into an executable, falsifiable research programme.

Data and code availability

Code and machine-readable artefacts are available at https://github.com/RAPIDENN/HOLO_runner and archived under [doi:10.5281/zenodo.18224589](https://doi.org/10.5281/zenodo.18224589). Certificates and exact controls reside in `first_principles_audit/`. Third-party data retain their original distribution policies; the bundle records derived metrics and provenance, not restricted measurements.

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